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TOPICAL REVIEW

Digital Restoration of Cultural Heritage With Data-Driven Computing: A Survey

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ABSTRACT Digitized methodologies in the recent era contribute to various fields of automation that used to hold different interests and meanings of human life. Buildings with historical significance, cultural values, and beliefs are becoming an interdisciplinary field of interest, engaging more computer scientists nowadays. Such structures need more attention towards reconstructing their values using a flavor of computerized tools instead of brickwork directly. Due to the wear of time, the tiles and engravings of most of the historical monuments are on the verge of ruin, endangering significant historical values. In this survey, we rebuild the values by delving deep into the device and methodologies by providing a comprehensive understanding of emerging fields and some experimental decisions. We discuss heritage restoration from some essential papers on 3D reconstruction, image inpainting, IoT-based methods, genetic algorithms, and image processing. The survey explains Machine Learning, Deep Learning, and Computer Vision-based methods for various restoration tasks in the related field. We divide this into certain parts contributing to different fields that restore cultural heritage. Moreover, we infer that the techniques will be faster, cheaper, and more beneficial to the context of image reconstruction in the near future.

INDEX TERMS Cultural heritage, 3D reconstruction, classification, generative adversarial network, building information modeling, inpainting.

I. INTRODUCTION

The preservation of cultural heritages has been gaining considerable interest since the last decade. Buildings with cultural heritages are often seen with rich designs and imprints carved into the fences within a tile-like architecture. Some establishments have their design significance having self-reliant architecture. Among these, the Timurid architecture of Central Asia [1], situated in Samarkhand, Uzbekistan remains one of the brightest examples. Timur, during the period of his

dynasty, built eclectic mosques, madrasahs, tomb complexes, and mausoleums while seizing the lands of Iran, Iraq, Syria, north of the Black Sea, Anatolia, and the area of the silk road linking China and Europe [2]. The Temple of Apollo at Syracuse is carved with large missives into the top tints, light on the newborn greek colony [3]. Artistic heritage also includes statues such as the Holy Christ of Blood which is one of the most mystical images in Spain's history [4] and the Neptune statue in Bologna [5]. This polychrome wood architecture of Holy Christ of Blood is built in the Baroque period and holds a deep archaeological and iconographic importance. On the other hand, the Fountain of Neptune (the

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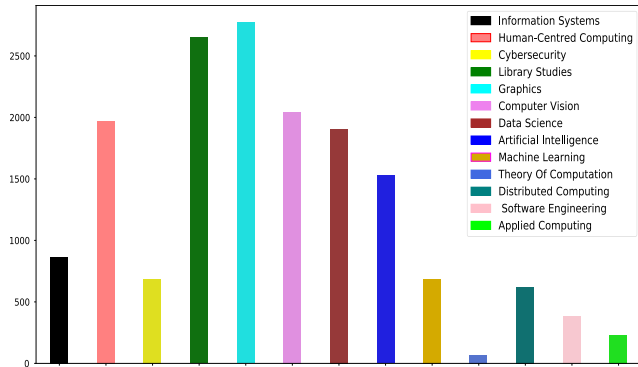


FIGURE 1. A synopsis of the current research varieties in computer science confronting research works in heritage restoration. Such works are contemporary with the current interest in working as a computer scientist from an algorithmic standpoint to the task rather than brickwork.

roman god of water, seas, and storms) was built in 1565 to renounce the power of the new Pope Pius IV. As we move towards the digital future of the materialistic world, historic buildings remain unobtrusive, losing their cultural significance through time. Heritage restoration is unfolding several emerging sites of Computer Science engaging more computer scientists nowadays is references by Figure 1 collected from web source.¹

In this survey, we come across several heritage restoration attempts that might be useful. Figure 2 diagrammatically represents the various portions discussed in the article. Apart from the historical significance of heritage architecture, we begin with diverse data collection approaches using several computerized methods. One is photogrammetry(PH) [6], [7], which is significant for the 3D reconstruction of images clutched from different angles, calibrations, and lighting conditions. We explore our sweep in this topic, mentioning varieties of PH methods considering the area of the space user wants to wrap. Thereupon, we depict various case studies of research papers such as Victoria Square [8], 3D scanning of Timurid architectures of Central Asia [1], [9], [10], Holy Child of Blood [4], [11]. Target and time-based capturing of the subject image, often termed Rephotography, is also a paramount factor in image restoration by analyzing the corrosion rate of the structure timely. Flaking of Magnesium Limestone in various heritage sites [12] and coal mining [13] are some popular applications of Rephotography. This method needs time-dependent images; thus, crowdsourcing of data [14] can be a beautiful resource allocation graph limiting the project's cost. Considering the cost-cutting as an effective parameter, Rephotography on a single image is accomplished by [15] with different lighting circumstances and scenarios.

3D reconstruction of collected data is a computerized method rather than mechanics, thus involving two primary investigation categories.

- 1) Image processing approaches extract influential features from 2D images to plot them into 3D images using kernel-based filtering methods and shape repair to reassemble the spot [16]. This variety of methods demonstrates the transformation as an orbit of the X-Y axis updating two-dimensional euclidean space compared to the Z axis to estimate a 3D transformation. Microsoft Kinect and Intel RealSense camera F200 are employed in Brazillian Baroque Sculptures to develop a Low-cost RGB-D sensor [17] using a pairwise alignment elaboration graph (Pulli's Algorithm). 3D reconstruction using Close Range photogrammetry techniques aims to extract adequate details from the 2D images. An equivalent approach using the A-KAZE algorithm is applied in five temples of Yogyakarta [18]. Extracting spatial and temporal knowledge from heritage architecture is a familiar way to rebuild a 3D model inferring Semantic Modeling [19]. Such methods engage CIDOC CRM to restrict knowledge representation graph [20].
- 2) Learning-based restoration techniques are time-consuming but exhibit better results than previous approaches. We start the discussion with the Cost Effective, High-Quality reconstruction of the CEPRO-QHA project [21]. The suggested method uses several open reference databases [22], [23] depending on the project necessities that match the data, followed by an annotation. During the model invention phase, the paper introduces a multimodal CNN classification model followed by Generative Adversarial Network [24]. SimpleRecon [25] turns out an energetic 3D reconstruction movement without 3D convolution directly towards 2D mapping. Another UNESCO world heritage site, Borobudur Relief, opted for guileless computerized methods for 3D reconstruction to construct a digital data library. The paper proposes the resolution by detecting soft edges heeded by densenet [26] and Atrous spatial pyramid pooling [27] layer. The paper prunes to capture soft detected edges from the consequential convolutional matrix to restrict the effect of preprocessing on Borobudur relief data. Furthermore, a method for reconstructing a forbidden city using GauGAN [28] is proposed. We examine a different section of GAN-based 3D reconstruction [24], [29], [30], [31], [32] in assorted heritage restoration tasks.

The article further discusses the idea of digital inpainting [33], [34] to search for holes in heritage architectures. The inpainting method uses several image processing techniques that entitle geometric characteristics of an object to fill up holes in a particular region [35]. The architecture of Hampi is exceptionally famous for advancing digital inpainting works via "Hampi goes Digital" [36]. [37] added a user venture into it without any prior knowledge of computer graphics. Such methods introduce 3D scan-based restoration [38] using digital re-colourization, photograph replacement [39]

¹<https://app.dimensions.ai/>

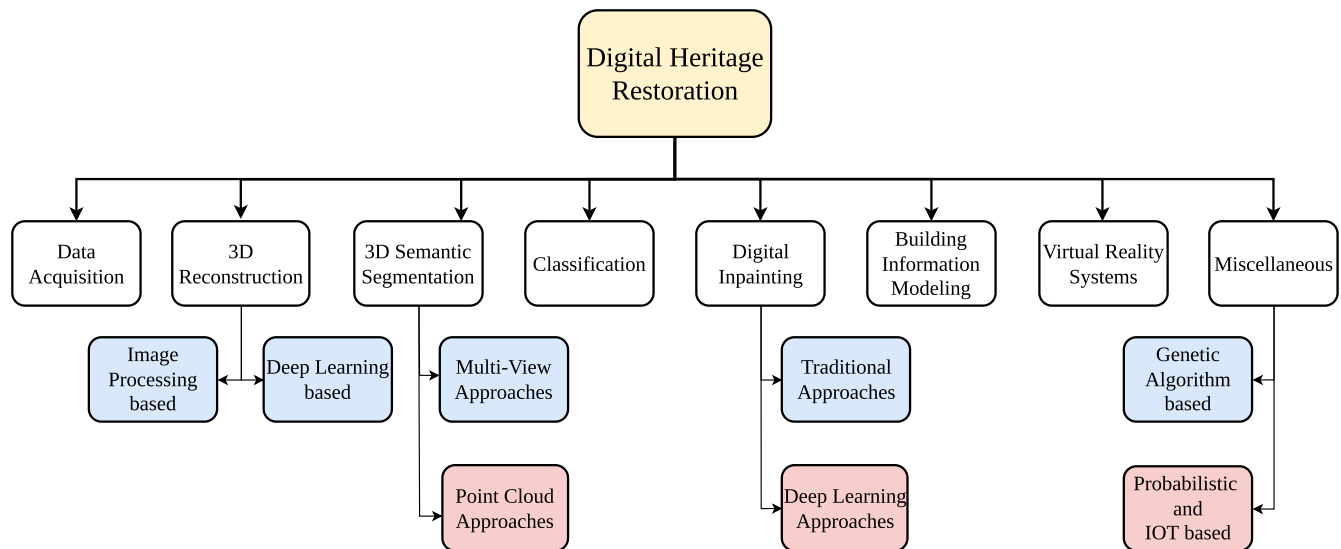


FIGURE 2. We present a pictorial recap of the subtopics we will discuss in the section on data collection. We unfold such cases concentrating on the stance of Computerized and Mechanical methodologies. Our focus areas predominantly include Data collection strategies followed by different restoration methods. Furthermore, the restoration incorporates 3D reconstruction, Inpainting, Segmentation, and approaches with discriminative procedure flows until the miscellaneous part engages state-of-the-art statistical approaches.

using affine transformation. With giant-lord DL advancement, many digital inpainting techniques transformed into GAN-based digital inpainting methods [40], [41].

Furthermore, to fulfill the anticipation of Machine Learning (ML) and DL models, [42] introduces a large-scale database. Image inpainting is the method for filling up holes or missing regions of an image. Inpainting can be categorized into conventional strategies and DL techniques. Ancestral methods either fill up missing parts with generated local structures or fill the missing portion pixel-wise, maintaining consistency with the surrounding pixels. These methods are suitable for filling up small missing regions in an image. However, the classic methods cannot handle image inpainting for large missing regions. Deep-learning-based inpainting techniques can suffice more extensive complex areas across an image. However, these techniques consume a large amount of memory and have high computational complexity. Deep-learning approaches may or may not be based on generative adversarial networks [43]. Salem in [44] presented a survey on various definitive and deep-based inpainting techniques.

Point cloud distinction using DL [45] instigate various Multi-view [46], [47] and point cloud based approaches [42], [48], [49]. A large-scale point cloud approach is influential as the DL-based tasks are most efficient on smaller data regions [50]. On the other hand, Dense Image Matching (DIM) [51], [52], [53] is used to produce dense 3D point clouds at a reasonable cost. It is a crucial piece of equipment for gathering three-dimensional point cloud data. Semi-global matching is one of the finest tradeoffs between matching precision and time cost (SGM) [54]. Meshes and textured meshes can be made using point clouds produced by DIM. A textured mesh adds colour information to the mesh by mapping an image onto its surface. A mesh is a polygonal model

that depicts the geometry of an object. For processing DIM data, a variety of software alternatives are available [55], such as SURE technology, which enables the automatic generation of point clouds, accurate orthophotos, and textured meshes from photographs. Classification after semantic segmentation as an efficient approach is a perceivable task to determine the object's qualities and observe the shape resemblance between other architectures of cultural heritage sites [56]. Several projects work [57], [58] using deep classifiers [59], [60] and point cloud approaches [47], [61], [62] investigate various approaches on heritage architectures around the world. We explore some important datasets [63], [64], [65] to accelerate the effect of DL. The development surrogate in the form of ancient script classification such as Mayan [66], and Egyptian [67].

Building Information Modeling is another pioneering solution in the effort of digital heritage restoration. From the detailed first introduction in the mid-1980s [68], this method adds information-based building modelling by various approaches in computer vision and image processing. However, this procedure is expensive but demonstrates good results, including Scan-to-BIM [69], [70] information modelling [71]. A semiautomatic approach for rooftops is proposed in [72]. A significant object segmentation using [73], a comparative study between 3D model and SCAN2BIM also shows a significant refinement in 3D restoration [69]. Followed by the architectural differences in BIM, a plethora of Virtual Reality systems is investigated using IOT sensors and scanners [74], [75], [76]. We conclude our discussion with more aspects of revitalizing the heritage of popular architectures using various probabilistic methods, IOT-based AI techniques, and genetic algorithm [77], [78]. We show the materialized importance of heritage restoration tasks to

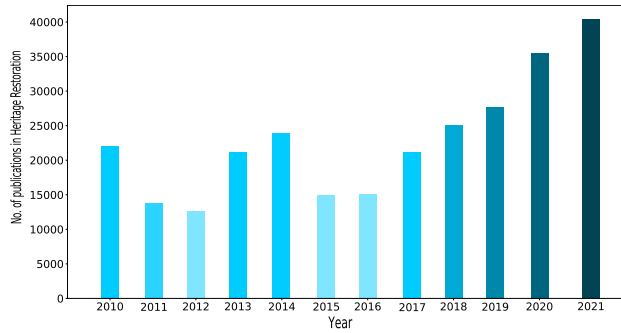


FIGURE 3. We demonstrate the significance of heritage restoration tasks in the last decade. The inclination of an increasing number of published research articles exhibits the growing inquisitiveness in this related domain.

support the growing interest within the community. Such a claim is supported by Figure 3.

The Organization of this article is as follows:

- 1) The importance of data acquisition and collection strategies, noting several approaches, is covered in section II. We begin with the fundamental photogrammetric methods and persist until visibility analysis constellates the photos pivoting on various complex parameters.
- 2) Section III takes the vision towards the effect of 3D reconstruction that inspires a better understanding of the heritage restoration task keeping a gaze on the advanced notions such as image processing and DL.
- 3) Heritage buildings are mostly disregarded, keeping the historical significance silent. Recoloring and brickwork without understanding the significance will be a massacre and thus need computerized tools in rescue. We illustrate section IV as inpainting algorithms that can either be applied or practical for heritage restoration tasks.
- 4) The article present numerous approaches of 3D restoration manoeuvring segmentation techniques in section V. We split the strategies into multi-view and point cloud techniques on the pertinence mentioned in several articles.
- 5) Examining the terms of restoration and generation, discrimination between the subject and cover remains an unsolvable part of the survey. We exemplify section VI to discuss the classification of several objects present in the heritage sites.
- 6) Physical installation of sensors boosts Virtual Reality and Augmented Reality techniques in this study. Virtual reality system in section VIII elaborates some advancements using motion sensor technologies installed in various church and heritage sites.
- 7) We discuss some additional vital methods that have been applied to several research papers in section IX and conclude with some essential future works (section X).

II. DATA ACQUISITION FROM HERITAGE BUILDINGS

A. PHOTOGRAMMETRY AND 3D SCANNING

Photogrammetry (PH), “Die Photometrographie”, is a technique to collect reliable information about an object by measuring distinct features using recording, photo capturing, and electromagnetic radiant imagery [6], [7]. It is the science of making measurements from photographs. This technique involves unique measurements of real-life 2D images for digitized 3D paradigms. The following are the steps involved in photogrammetry-based data collection:

- **Image acquisition:** Photogrammetry relies on photographs to generate 3D models. The photographs should be taken using a calibrated camera and should overlap sufficiently to allow for reconstruction. According to [79], planning is critical to ensure that the survey area is divided into sections to ensure that all areas are captured by the camera.
- **Image preprocessing:** The photographs should be pre-processed to remove any distortions that may affect the accuracy of the final 3D model.
- **Image matching:** The photographs should be matched to identify common features and to generate correspondences between them. This step is crucial in generating accurate 3D models.
- **Point cloud generation:** Once the correspondences are identified, a point cloud [80], [81] is generated using triangulation. This point cloud represents the 3D coordinates of the corresponding features in the photographs.
- **Surface generation:** A surface is generated from the point cloud using various techniques, such as interpolation, meshing, or contouring.

B. LASER SCANNING

Laser scanning [82], [83] is a non-contact method of collecting 3D data using laser light. The technology known as laser scanning, commonly referred to as LiDAR (Light Detection and Ranging), employs lasers to collect information about the size and shape of objects and settings. Map-making, surveying, and the creation of 3D models of things and locations are all common uses for this technology. The inspected region can subsequently be represented in 3D or on a map using these point clouds. The following are the steps involved in laser scanning-based data collection:

- **Scanner setup:** The laser scanner is set up in a position from which it can capture the desired area. The scanner parameters, such as resolution and range, are set according to the required level of detail.
- **Laser scanning:** The laser scanner emits laser light that is directed towards the target. The laser light reflects off the target and is captured by the scanner. The scanner measures the time it takes for the laser light to reflect back to the scanner, which is used to calculate the distance to the target.
- **Point cloud generation:** The scanner generates a point cloud [81] that represents the 3D coordinates of the tar-

get. The point cloud can be visualized and manipulated using specialized software.

- **Data processing:** The point cloud is processed to remove any noise or outliers that may affect the accuracy of the final 3D model.
- **Model generation:** A 3D model is generated from the point cloud using various techniques, such as meshing, contouring, or voxelization.

There are various kinds of laser scanners, including ground-based, aerial, and mobile models. Terrestrial scanners are used to detect modest to medium-sized items and settings, and they are often tripod-mounted. Large areas, such woods or coasts, can be scanned using airborne scanners, which are placed aboard aircraft. Roads, bridges, and other forms of transportation infrastructure are scanned using mobile scanners that are placed on moving vehicles. With many variants (section II-B), photogrammetry and laser scanning is one of the most crucial techniques to collect photos for further consideration in digitizing cultural heritages.

While digital image capturing consists of different stages of photo capturing, PH extracts as many features as it can extract from a 3D structure by capturing photos from different angles, lighting conditions, and other aspects.

- **Terrestrial:** This approach [84] motivates capturing photos from one location on air or ground with a hand-held camera or tripod.
- **Aerial:** This type of PH covers a much larger area than the previous one [85]. It is mainly captured from a flying aircraft (e.g., Satellite imagery) or a camera in the air [86]. Aerial imagery can further be divided into vertical and oblique PH based on the axis of the camera. While many cameras are available in the market, this method includes lightweight ones for capturing meaningful photographs.
- **Close Range:** The range is an important aspect while capturing the images. While Aerial photography covers more area capturing the image from a far distance, close-range photogrammetry [87], [88], [89] considers taking the images from a lesser range from the terrain. This technique is useful when preserving minute details in the architecture rather than covering more parts.
- **Stereophotogrammetry:** This special PH method involves determining 3D coordinates from one or multiple 2D image coordinates taken from different positions [90].

Starting with the idea of physical model creation from 2D images nowadays motivates a plethora of software that can work with 3D reconstruction from 2D images. The work of 2D image capturing includes photogrammetric approaches in various cultural heritage sites and architectures. This method manipulates the camera parameters (focal length, adjusting lighting conditions, principal component), resulting in non-linear solutions to extracting homologous coordinates [91].

C. VRML MODEL OF VICTORIA SQUARE

[8] refers to various laser scanning techniques on Victoria Square in Gorizia, Italy, to reconstruct a 3D city model. The efficiency of aerial laser scanning allows the acquisition of millions of 3D coordinates by emitting higher laser beams to the subjected architecture. The method also includes an effective segmentation algorithm by filtering Digital Terrain Model (DTM) from Digital Surface Model (DSM) with additional condition variables like Digital Elevation Model (DEM). Terrestrial Laser based PH constructs the image with a much greater level of detail by scanning the architecture from different static locations.

D. 3D SCANNING OF TIMURID ARCHITECTURES

In the expedition to Central Asia (CA), the Department of Computer Science of Lublin University of Technology (LUT) carried out multiple efforts to use modern technologies for cultural heritage restoration, and preservation [1].

- 1) In the first stage, remote communication technologies are used to collect 2D photographs engaging museum workers to make a 3D model. The PH technique focuses on collecting terrestrial photogrammetry covering minute details of the architecture.
- 2) In 2017, a collaborative effort between LUT-CA Corporation was carried out using 3D scanning of large monuments and terrain structures.
- 3) The committee has carried out multiple expeditions based on the requirement of 3D modelling enforcing 3D computer scanning in Samarkhand, development of templates for the renovation, analysis of surface tile colours on Sherdar Madrasah Mosiac [1], launched “3D Digital Silk Road” internet portal covering some of the popular Timurid architectures.
- 4) The committee further aims to cover 3D scanning of petroglyphs and more minor artifacts from the National Museum in Uzbekistan.

Focusing on the data collection stage and the professional documentation of historical artifacts in these areas, PH scanning can be replaced using terrestrial laser scanning technologies. These are commonly used for 3D modelling forming image-based registration (IBR) [9]. However, the time-consuming nature of laser scanning often fails to identify crucial parts of minute details of the architecture sometimes. Thus, enforcement of a new method, SFM (Structure from Motion) [96], [97], is a PH-based approach; placing as many points as possible on the architecture is crucial. Using a series of 2D images or video frames, the SFM method can be used to reconstruct a 3D model of a scene or an entity. There are three basic steps in the procedure:

- **Feature Extraction:** The SFM algorithm retrieves and extracts distinctive features from the images, such as corners or edges, at this step. The identical pixels from different photos are then matched using these attributes.
- **Camera pose estimation:** The SFM algorithm calculates the camera pose and intrinsics (such as focal length

TABLE 1. We have accumulated several notable methods for heritage data collection and detailed those in Section II.

Author	Year	Comments
Rosten et al. [92]	2006	An efficient ML approach for high-speed corner detection irrespective of frame rate drop is helpful for faster applications. The exploration proposes 3D feature-based learning for the reconstruction task engaged in heritage restoration
Vistini et al. [8]	2007	An integrated semi-automatic method of photographic photos for the task of virtual rendering model of Victoria Square in Gorizia (Italy)
Doehne et al. [12]	2008	A time-dependent, power-efficient depth estimation camera is developed to analyze pattern and rate of synchronized magnesium stone loss due to environmental conditions
Schreiber et al. [93]	2012	Newly introduced, 3D user scanner-based approach to execute Romanian parts in the underground museum of Cologne
Evin et al. [94]	2016	A close-range photogrammetry technique delegated by a high-end surface scanner is used to reconstruct five wolves of crania. The geometric comparison is calculated via a deviation map that takes 3D landmark-based morphological analysis to complete the task
Shishido et al. [14]	2019	An improved method of data collection is introduced by crowdsourcing photos from cultural heritage sites followed by an autoencoder-based feature retrieval to transform it into 50 to 100-year-old photo
Milosz et al. [1]	2021	A joint operation between LUT and CA team for expeditions through 3D scanning-based approaches on Timurid architectures
Police et al. [95]	2021	Lighting conditions often work as a barrier in the data acquisition stage, maintaining it for further processing. This paper proposes a user target-based visibility study that nullifies the effect of different environmental conditions on the user's perspective
Ruiz et al. [4]	2022	The digital twin-based 3D scanning approach by lessening aerial photography followed by digitized technological advancements

and distortion parameters) for each image once the features have been recognised and matched. The direction of the camera and position in relation to the 3D surroundings is referred to as the camera posture.

- **3D reconstruction:** After estimating the camera poses, the SfM algorithm triangulates the feature points' 3D coordinates by intersecting the camera rays from several viewpoints. The resulting 3D point cloud is sparse and can be improved and densified using methods like bundle correction and dense reconstruction.

A revisited version of incremental SFM is proposed in [97], that includes robustness, accuracy using geometric verification and image matching.

The specialists applied using Trimble DR 200 Plus 5602 device to enforce the SFM method to create the 3D models of megalithic monuments located in Vales [10].

E. DATA PROCESSING OF THE HOLY CHILD OF BLOOD

[4] discusses preserving the masterpiece, Holy Child of Blood. During the *Spanish Civil War*, most of the parts of the architecture were destroyed; thus, it needs restoration using recent technological advancements. However, state-of-the-art methods involve metric evaluation on heritage assets [98], structured light hand scanner [99], [100], SFM Software [11], and many more, they use ultraviolet light to study the alteration of the actual and restored one. The architecture creator, Nicolás de Bussy, used flash paint, including a single layer of original paint. We illustrate some of the most commonly used digitization techniques for heritage sculptures. Digital PH is one of the essential methods but fails due to low-light conditions and dark surfaces, whereas 3D scanning is much more challenging to handle. We infer the paper applies photogrammetry, laser scanner, and structured light scanner-based approaches to the Holy Child of Blood. The

structured light scanner projects white or blue light on objects rather than the laser scanner beams. It calculates the distance between points from the field of view resulting in an effective method with faster, more accurate, and high-resolution determination [101].

F. OTHER METHODS

A plethora of methods can be applied using photogrammetry and 3D scanning technique to contribute to the field of cultural heritage. [93] formulates a phase-based 3D laser scanner for quantitative damage analysis during the excavation of construction of the historic city centre of Cologne. The article presents documentation of restoration works that need to be done to avoid the collapse of the construction site. In the context of large-scale data collection, [10] introduces both virtual and tangible settings as a touch table to examine data from heritage sites using alternative representations. The method engages extensive applications from heritage data for public record educational and research methods in the near future. We refer to [102] that expands the interest in 3D photogrammetry in terms of functionality and portability. It is suitable and cost-effective compared to 3D imaging and scanning techniques by putting geological remote sensing devices near the physical object. A close-range photogrammetry mechanism has been discussed in [94], keeping the scanner and 3D-based modelling out of the concept for less computational performance.

G. TIME LAPSE IMAGING AND REPHOTOGRAPHY

Time Lapse Imaging refers to a special type of rephotography of a particular object in a specific period. For digital heritage data collection, Time-lapse imaging has been applied to analyze the loss of stone curvatures in various architectures and monuments. This type of rephotography also motivates

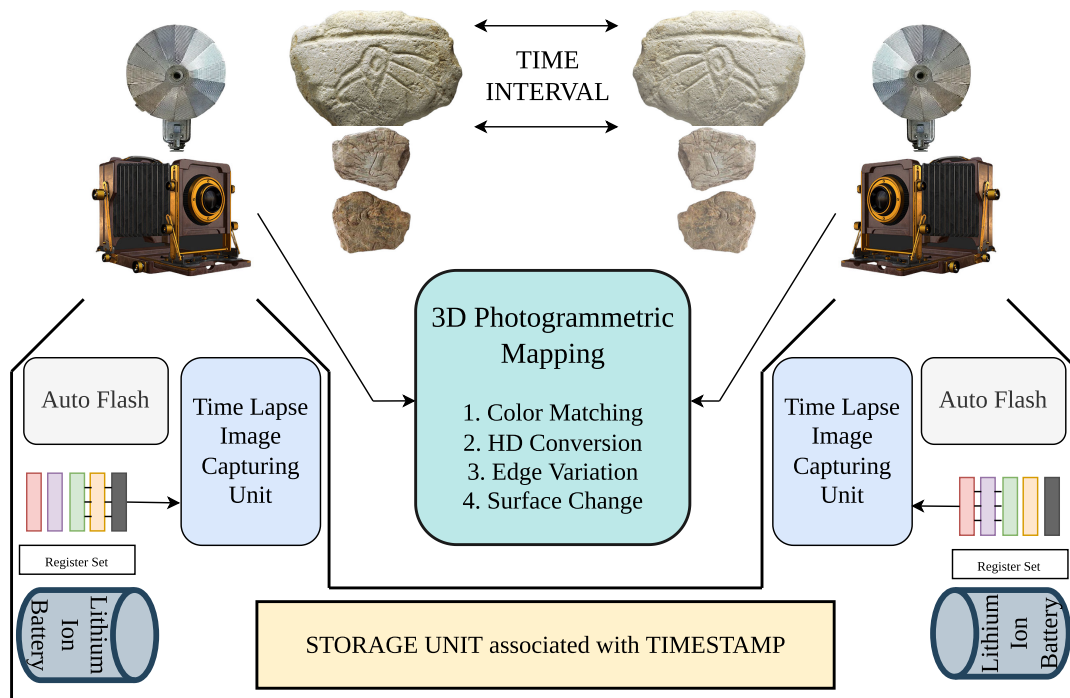


FIGURE 4. We illustrate the methodology mentioned in [12]. The process engages Time Lapse Rephotography and thus serves an interdisciplinary approach between mechanics and computer science. The structure of the Camera is updated based on the requirement. At the same time, the time interval mapping matches the flaking of limestones based on algorithms to calculate the decay rate. We present a graphical mapping between time interval photography integrated with digital circuitry in this task.

the physical restoration of terrains suggesting the reasons behind the flaking of particular areas. It enforces various types of photography while capturing the subject at a specific time interval. Ultraviolet photography, X-ray radiography, and infrared photography are applied to detect the damaged region that is not visible to the human visual system. Time-lapse imaging has already been used in different fields of coal mining [13] to understand the location and depth of exploring old gallery mines. [103] experiments prokaryotic cell track analysis based on segmentation. This particular type of imaging is essential for a fast-track video that can present episodal changes in the object visible to the human.

1) ANALYSIS OF FLAKING OF MAGNESIUM LIMESTONE

[12] presents a time-lapse imaging approach capturing a series of images in regular intervals to experiment with the decay of magnesium limestones in various heritage sites. This indicates episodic loss of stones due to environmental conditions such as high winds or synchronous for all nearby stone blocks for surface changing. Historians often applied these methods to study wall paintings in the mid-90s. The authors developed an updated version of time-lapse photography using terrain structure analysis with visual ground conditions and macro imaging with an interval timer. The time-lapse field imaging consists of a standard seven-pixel camera having 4GB of memory. They applied a cost-effective, easy, and time-consuming method to study the rapid stone loss

from a stairwell nearby. The cameras are manufactured with time-lapse circuitry connected with an acid battery to capture full-resolution photos every three months for 3 hours on a lithium-ion battery. Figure 4 presents a visual understanding of the time-lapse imaging capability used in the paper. For further processing of the data, three steps are involved:

- Color correction using *Match Color* command in *Adobe Photoshop* to normalize some exposure and lighting conditions.
- The images are further compiled into an HD video format using Apple Quicktime software to verify a small number of losses visually.
- Different error conditions, such as slight surface change and edge variations, are quantified using Photoshop.

2) CROWDSOURCING OF LONG INTERVAL TIME LAPSE IMAGE

The importance of long interval time-lapse imaging is more important than the collecting process mentioned in [12]. The preservation of heritage buildings is extensively dependent on older data where more architectural richness is present. To this effort, [14] enforces crowdsourcing as a crucial component. The paper promotes four types of crowdsourcing to preserve the cultural heritage site.

- 1) Public crowdsourcing: Image capturing tasks were outsourced to unspecified visitors and workers around

the internet in a specified manner that matches the alignment, position, and orientation of the same terrain image captured over 50-100 years ago.

- 2) **Image Shooting:** The community has been given extensive permission to use various consumer products such as cameras, stages, and platforms near the terrain structure to ensure the quality of the image. The committee sets up a cloud server where the user can upload captured photos.
- 3) **Generation of Time-Lapse Video:** A computerized DL method, Autoencoder [104], is introduced to generate a time-lapse video using the image matching technique [105].
- 4) **Expert Level Crowdsourcing:** This part engages archaeologists to analyze the time-lapse video by observing changes. Certain calamities should be replaced; thus, they provide the parts of the old photos that need to be restored within the newly captured image.

The procedure implementation results in high-resolution restored images with partial black shading parts, indicating the decay rate in the newly captured images.

3) HIGH SPEED CORNER DETECTION

While working with the 3D reconstruction of architecture using different approaches, corner detection is often costly and results in defective reconstruction. Thus, corner detection in prior can be an efficient method to enforce effective data collection. [92] refers to an ML-based technique for high-speed corner detection rather than FAST, non-maximal suppression. The paper defines CCN (Consistency of Corner Numbers) and accuracy as effectivity parameters for the repeatability of corners. The margin of error should be considered due to the imperfect alignment of the physical image capturing and maximum point detection on a slightly different part, assuming the scenery is new.

4) REPHOTOGRAPHY ON SINGLE IMAGE

Rephotography is one of the methods to measure visual and social change around a structure captured from several positions. A photographic photo is the best way to analyze visually attractive changes. Allowing time as a changeable parameter, rephotography can analyze changes in architecture [15] during the period. Repeating photograph [106], Rubber sheeting [107], [108] is an old yet effective time-consuming photographic method. Computational Rephotography assisted with ML is getting more attention due to its more superficial, time-effective nature. [109], [110], [111] presents partial automation in computation rephotography using multiple sequences of images. However, techniques such as Structure from Motion [112], Multiview Construction [113], and Image-based Rendering [114] are essential to detect camera position for better photography depending on the various light conditions.

H. VISIBILITY ANALYSIS

Visibility analysis determines what can be viewed by an observer from a given location. We can make predictions regarding the depth of a single image under different lighting. The limitations of such predictions have been discussed by [115]. Further, a new approach has been proposed to reduce the images' shadows and reflections. Here, images of ancient Roman coins under various lighting conditions have been used as a training dataset. Visual Acuity is the clarity of vision at a given distance. Visualization of heritage sites and artifacts based on visual acuity is demonstrated through the iconography of the Cathedral of St. John the Theologian in Nicosia, Cyprus, in [95]. Raman spectroscopy is a spectroscopic technique that relies upon Raman scattering. Casadio et al. discuss the frontiers of the application of Raman spectroscopy, such as detection, analysis, and ageing monitoring of archaeological artifacts in [116]. Further, Lazzari and Reggio in [117] demonstrate the applicability of surface-enhanced Raman spectroscopy (SERS) for artworks.

III. 3D RECONSTRUCTION: IMAGE PROCESSING AND LEARNING BASED APPROACH

The practical intuition behind Image Processing works with images to transform them into digitalized versions along with multiple filtering and morphological operations based on the user's need. It focuses on noise reduction, changing light conditions, and many more procedures to extract meaningful features from existing images instead of directly using rephotography. On the other hand, ML is a fashionable way to solve a problem in the recent era. It represents the machine-level decision-making capability instead of statistical methods to train such learning models by loading data with excellent parameter settings. The model learns the patterns inside it and, given a test environment, decides. Rather than looking into the mathematical and theoretical intuition of image processing and ML methods, we expand our article toward the direction of 3D Reconstruction. We present this technique in both image processing and ML-based approaches. However, the power of both courses has been debated in the research community for decades until learning-based designs exhibit better applicability in various fields with recent advancements in computational capabilities.

The physical structure of the real-time three-dimensional objects is delusional by the 2D impression captured by a standard digital camera. The 2D coordinates depend upon the X - Y plane, which explains that any point (x1, y1) is x1 unit distance in the positive X axis and y1 unit distance in the positive Y axis. The distance between any two points (x1, y1) and (x2, y2) can be defined as

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \quad (1)$$

The representation of the 3D coordinate includes depth mapping [118], [119], approximation with numerical approaches [120] to estimate the 3D profile along with the X - Y axis such as Z. It is a process to create a virtual 3D

system with the help of techniques presented in the section II with a particular system environment. Firstly, The model captures the position of the distinct target object's negative or mirror image. This step has been taken by the perspective projection method inferring from the study of optics. The distance between the target and the image is based on the principal component distance from the centre of the image plane [121]. Secondly, such a mirror image needs a positive image to estimate the 3D depiction of the central axis from the sets of images. Thirdly, the rotation of the (X, Y, Z) coordinate is represented using three ordered rotations of the X , Y , and Z axis consecutively that can be defined as $(\omega, \varphi, \kappa)$.

Heritage buildings are often dilapidated and need further attention. With the greater computational power, we apply 3D reconstruction. Thus, we want to discuss 3D reconstruction as an essential parameter for digital reconstruction in cultural heritage. Digital technologies are considered to be a reliable mode for the preservation of cultural heritage. Several ML and DL approaches are used. Belhi et al. in [21] investigated several DL approaches for classifying and annotating heritage data and filling the portions of missing data.

A. IMAGE PROCESSING BASED APPROACHES

1) LOW-COST RGB-D SENSORS ON BRAZILLIAN BAROQUES

The motivation behind this article is to improve a low-cost solution using RGB Depth sensor cameras on famous sculptures such as Buddha and Mozart. [17] experiments with two types of 3D sensors, Microsoft Kinect and Intel RealSense Camera F200, that use structured light technology. MS Kinect is faster and thus helps real-time performance. On the other hand, Intel RealSense uses the same technology with a different set of lenses and general specifications. The stages performed (Figure 5) for the method are defined as follows:

- This stage scans the structure's surface area using [122]. The sensors have been set to the natural option time to higher but not the highest resolution photos for 3D reconstruction. This type of real-time reconstruction is dedicated to the user's visualizing of the structure in real-time but not with the best quality. They set the picture quality to 640×480 with N number of coloured images of RGB colour range in the exact resolution. The image contains a 1mm error when taken from a distance of 1m.

- For efficient noise reduction, [123] apply a particular type of weight joint bilateral or refinement filter for depth maps. The filter applies a jointly computed kernel using convolution operation to a raw image. Furthermore, it applies a sloping depth computational filter to remove blurriness accessing the boundary region. Additionally, keeping the computational cost low, it provides an extensive range of filtering approaches in the case of depth maps, such as block-matching and Markov random field methodologies. The weighted average of the neighbouring pixels calculates the depth value of each pixel. The base point's gaussian distribution is updated based on the gaussian curve present in the nearest ones. However,

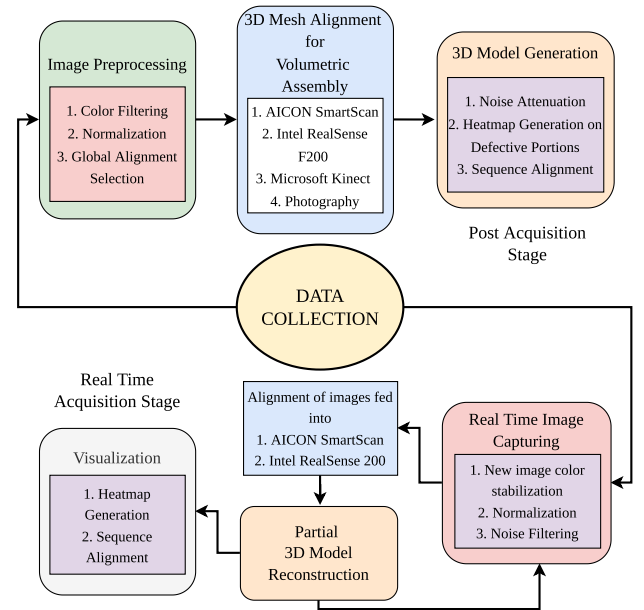


FIGURE 5. The workflow of the low-cost RGB-D sensor introduces a two-stage method for the reconstruction task. The post-acquisition stage engages captured images and results in a fully reconstructed structure based on the volumetric assembly stage performed by different camera devices. On the other hand, real-time acquisition partially reconstructs the data based on noise filtering and engaging camera device photos efficiently.

the feature similarity metric from the particular base point can be a valid reason for spatial and temporal information that covers other aspects like lighting conditions, mesh projections, illuminations, and many more. Therefore, further updates in MsKinect, such as bilinear upsampling of the images, could maintain higher resolution photos giving a better user experience discussed in [124] in a controlled way.

- After completing the image acquisition and processing, the paper further investigated the pairwise alignment of the images by converting the data into Global Coordinate System (GCS). [125] proposes one of the most popular ICP (Iterative Closest Point) algorithms for better geometrical alignment of 3D models when the base point is known. However, it flats mostly in meshes with small feature space and less similarity; therefore, many ICP methods have been proposed lately. This method is faster and considerably performs, assuming an excellent initial base model. It has become a dominant method in the field of 3D alignment and thus has central importance in this study. [126] presents the previous one along with the acceptable registration of covered pictures in case of fusing 3D surfaces. Based on the ICP, the proposed pairwise alignment covers mostly similar images (95% similarity) to converge different points at a 30fps rate.

- Pulli's algorithm scans data pairwise to use the pairwise alignments as a constraint to generate the registration errors. Overlapping surface sections are introduced to minimize the chance of getting stuck into a local minimum. This expansion

is better than applying rephotography as it works for large datasets and generates pairwise empirical results from spatial features to generate excellent alignment of the data pairs.

2) 3D RECONSTRUCTION USING CLOSE-RANGE PHOTOGRAMMETRY IN YOGYAKARTA

Unique regions in Yogyakarta (situated in Indonesia) have been ignored over decades, thus containing a large amount of structural damage due to unavoidable natural calamities. The event resulted in the degradation of the region's spiritual and cultural heritage aspects, thus requiring special attention using computerized methodologies. However there are plenty of 2D photos available around the globe, but they usually lack in-depth information regarding the particular object. [18] refers to a method of close-range photography to capture minute details of heritage structures.

- The committee collected 1319 photos by visiting five temples in Yogyakarta.
- Keypoint extraction from the collected images using the A-KAZE algorithm [127] is carried out, which is a faster version of Modified Local Difference Binary Descriptor for highly efficient gradient information technique.
- Additionally, creation of a 3D reconstruction model after A-KAZE algorithmmakazepablo2013 keypoint extraction using LIOP (Local Intensity Order Pattern) [128] as a feature matching technique is introduced, followed by the creation of Sparse Point Clouds, referring [80].
- Furthermore, the paper demonstrates FSSR (Floating Scale Surface Reconstruction) [129] and LST (Large Scale Texturing) [130] to deal with the shape of the surface and texture of the terrain using MVS (Multi-View Stereo) as a geometrical shape generator. [18] accumulates some important SOTA methods to reconstruct the heritage of Yogyakarta Temples by creating a higher resolution 3D reconstruction of the architecture.

Index Terms:

A-KAZE: A low resource, Non-Diffusion Filtering method using FED (Fast Explicit Diffusion) [131] for efficient feature detection. In this paper, the author proposes a method for updating the FED network and generating much faster and more accurately in a pyramidal framework for feature detection called Accelerated KAZE.

FSSR: Post-processing stage consists of multiple operations. One is the Floating Scale Surface Reconstruction (FSSR) method to generate geometrical structures within point clouds. It is a multiscale reconstruction method enabling finite surface area present from the large, noisy sets.

LST: The effort is to develop a texturing-based 3D reconstruction model independent of drastically varying properties within the images, such as blurring, lighting conditions, occlusion, and exposure variation. These properties do not impose the differentiation between the 3D reconstruction model and the same images and assure the preservation of the colour conditions [132].

3) SEMANTIC MODELING OF ANCIENT EGYPTIAN SITE

Tell el-Daba (TD) is a historical site near Egypt's eastern Nile Delta Region. Since 1966, the popularity of the particular region has increased due to multiple excavations and archaeological works. Various historical evidence from the early second Millenium BC is being recovered from the site. This site's popularity caused a shift from analog to digital technologies for advancing technology, thus engaging more computer scientists to generate empirically effective results. As a result, archaeological communities such as OREA (Institute of Oriental and European Archeology) focus their interest on the digital reconstruction of these sites to enhance historical importance. The researchers impose the methodologies, guidelines, and infrastructures under the ARIADNE (Advanced Research Infrastructure for Archeological Data Networking in Europe). The effort toward semantic modeling refers to the spatial and temporal information modeling on a virtual reconstruction TD model, enhancing the option of a prosperous semantic representation technique.

We infer the work of 3D reconstruction discussed in [19] as an advanced, well-documented, multimodal approach to emphasize the digital understanding of TD. CIDOC technology refers to the Centre for Intercultural Documentation, adding CRM into it, focusing on the Conceptual Reference Model for mediating varieties of heterogeneous data for cultural heritage. CIDOC CRM discusses a type of ontology [20] with a definite knowledge representation technique on cultural heritage data. This technology allows access to the core ontology of different places (FRBRoo, CRMSci, CRMarchaeo) of data collection, such as museums, libraries, and data archives of specialized cultural datasets. It defines an Entity Relationship Model based on the research-based understanding to create an object-oriented, semantically richer model to support a broader range of use cases. To convert back to ER from the CRM model, they need additional software support to restore semantic information. There are two types of CIDOC CRM [133] model representation; persistent modelling is represented by an indeterminate period, whereas, Temporal modeling is bounded by time definition. However, the model depends on linked data structure rather than nodes to sound the Entity Relationship(ER) Model more conceptually understandable. CIDOC CRM method has a considerable performance in the case of cultural heritage preservation. Generating an end-to-end model accumulating the set metrics along with the user-based data input remains with much larger commendable support to the user. Multiple steps are incorporated, which can be defined as follows:

- The process starts with excavating meaningful materials in the archaeological site. The researchers make a texture analysis graph model to create the report from the site with pen and paper. Furthermore, digitizing the handcrafted document has been put into a scanner and edited based on viable options for image processing approaches.
- After the document acquisition process, the paper describes the importance of class-based modeling for the CIDOC CRM technique. The classes further can be divided

into 1. physical Thing E18, 2. Activity E7, 3. Information Object E73. However, the class-based understanding varies from application to application; we infer that these three class weightings are primarily used for this purpose.

- The knowledge graph derives essential components in the workflow and creates connections via graph-based analysis. For this purpose, the graph focuses on creating analog documents with the documented ones with 1 million triplets collected as the comprehensive metadata. Understanding the usage of resources for different structures, the historical sites that need to be documented, and many more aspects are essential. The knowledge graph provides a clear understanding of all the contents and connection queries made by the practitioners.

- Finally, the 3D reconstruction model has been created based on the previous stages. It contains three classes defined in the documentation process with mid-connections and data-sharing techniques based on the knowledge graph representation. The archaeological unit containing the finding has been accumulated to the augmentations stage via a P16 which helps the reconstruction process further in the Activity E5 stage. Information object class modeling is provided from the previous two operations, analyzed by P92 and P70, including the source and interpretations made from the augmentation and reconstruction to generate the 3D models in the D1 stage.

4) A COMPLETE SYSTEM FROM REASSEMBLY TO SHAPE RESTORATION

Physical restoration is a complex, time-consuming job given a large amount of data with large sparsity in the data space. Thus, the digital restoration process is essentially based on the virtual reconstruction of the specialized object-type models. Furthermore, the shape is another important aspect of the objects in heritage structures. Digital shape restoration is a vital subpart of object reconstruction, considering the fine-tuning of the model based on shape and structure analysis. [16] proposes an end-to-end pipeline for shape reassembly from fragment mesh analysis to reconstruct missing parts based on the complementary and repaired geometric aspect of a 3D object. Firstly, the process collects salient and fracture surface features from the intact side of the heavily damaged architectures. Secondly, the process continues with the concept of generalized symmetries side by side with complementary part extraction focusing on the importance of the subpart. There are multiple parts to the actual process, which are defined as follows:

- The step applies for surface registration in potentially fractured parts, excluding the intact part of the process. These feature curves are maintained using the extrapolation curves closely following the trajectories.

- This part is the combination of three subparts. **Pairwise Optimization** inputs the preprocessed image and matches it with the closest fragment set variety. Additionally, the inclusion of feature curves is optional based on feature experimentation. **Multi part repairment** sets the aim to the multi-part restoration process after the single-part restoration. Inter-

part penetration process repairs the connections between multiple fragment sets divided by the age of restoration. **Reassembly** of these various fragments is initiated by the user when multi-part refinement is complete. The clusters have the architectural result of the initial portions, given that reassembly will be targeted to a distribution based on the user's choice.

- The pipeline uses VRMW software to reassemble the parts. **Object completion** is done by symmetry analysis from candidate pictures followed by a computation of self-registration of the incomplete subpart. A **merging and annotation based** process is added to identify the complete subpart geometrically. This annotation analyzes the following method of **missing part extraction**.

- Generation of the new image needs further preprocessing and pairwise alignment to detect the correct position of the particular pixel values [134].

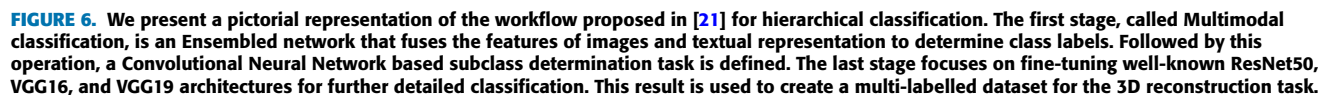
- This detailed step consists of expanding the clusters symmetrically to concise the planer, rotational, and transitional types of symmetry. The balance analysis is also done by plotting the heatmap to the reconstructed part using the surface function. The heatmap strictly shows the difference between the reconstructed and original parts of the object.

B. DEEP LEARNING BASED APPROACHES

1) DEEP LEARNING BASED RESTORATION ON CEPROQHA PROJECT

CEPROQHA Project refers to a Cost Effective, High-Quality solution for the Restoration of Qatar Cultural Heritage sites using Advanced cutting-edge Holoscopic 3D Imaging. Various approaches promote using AR and VR to design virtual museums for user experience. On the other hand, applying DL and ML is a fashionable trend nowadays, given the broadly available datasets. [21] proposes a DL-based solution for the CEPROQHA project, accumulating many datasets to restore, classify, and annotate enormous missing data, mapping standardized pieces of information in the cultural heritage sites. Recent advancements in Generative Adversarial networks (GAN) contribute to generating new image data and understanding the training dataset's pattern. These techniques are leveraged to enhance the task-based DL restoration techniques in cultural heritages [135], [136]. Figure 6 represents a simple process workflow for the DL-based restoration proposal of the CEPROQHA project. DL-based tasks can further be divided into discriminative classification and GAN-based restoration approaches proposed in [21]. The paper presents recent breakthroughs in DL-based rehabilitation for CEPROQHA projects encouraging transfer learning-based approaches with similar datasets in this domain. The key highlights of the pipeline are listed below:

- DL-based approaches feed on broadly available data. The authors participated data oriented cultural heritage challenges to work with available datasets similar to the project [23]. Furthermore, they have accumulated one of the most extensive



- After a particular data collection and preprocessing stage, two types of model creation stages are promoted. *Multimodal Classification* proposes a multimodal-based classification approach from painting images and their genre using ink and paper-based description. The model captures the feature from the painting images using a feature extractor of the Resnet50 [59] model. Subsequently, the texture features are

- The research further develops an end-to-end reconstruction system using Generative Adversarial Networks [43]. The underlying architecture of the study is the Deep Convolutional Generative Adversarial Network [24] to capture features from a set of images and generate fake photos to fool the discriminator. Using a clustering-based data discriminative approach, the pipeline trains several DCGANs based on the category of cultural heritages based on the last part. The generated data from a network using a classifier will further enhance the selection of the model to create specific and

detailed data. We can also apply multiple NLP techniques on ontology-based architecture reconstruction, given the solution using CIDOC CRM [133] architectures.

2) SimpleRecon: RECONSTRUCTION WITHOUT 3D CONVOLUTION

3D indoor reconstruction is a two-phase process including patch-based depth estimation accumulating depth merging with surface reconstruction. Both methods depend on 3D Convolution, limiting the real-time and computational applicability. SimpleRecon represents a carefully designed metadata-inclusive method that promotes a 2D convolution neural network feature map extractor. Integrating the keyframe and geometric data depth estimation allows an informative plane-based scoring. Given a limited memory, extracting 2D Convolution maps in a feature-based extractor on metadata description is a nice way for indoor scene reconstruction.

- Often, cultural heritage reconstruction depends on indoor scene restoration using 3D methods for study and architectural richness mapping. In this effort, indoor scene reconstruction is reasonably necessary for documentation purposes so that the building structure remains intact for decades for historical significance. ATLAS [137] is one of the previous attempts that use the Truncated Signed Distance Function (TSDF) to extract 4D feature volume from input images using expensive 3D convolutions. *Going back to the basics*, the proposed approach conducts a depth plane scoring using Multi-Layer Perceptron, followed by solid image metadata based on geometric loss.

- The model representation uses various referential source images as input. Based on [138], the features are extracted using a 2D convolution operation followed by a multilayer concatenation in cheaply available, carefully augmenting metadata. These metadata hold information about the 3D relationship between the referential images to build the loss value. Multiple processes such as *Feature based Dot Product*, *Reference Plane Depth*, *Depth validity mask* identification techniques are added via a multi-projection scheme. However, the features from metadata are further passed into the MLP to weigh the contribution and extract valuable information from each pixel rather than a convolution-based operation. The model is trained on a 3D reconstruction dataset ScanNetv2 [139] that comprises 1201 training, 312 validation, and 100 testing data without a fine-tuning method on seven scenes.

- The supervision of the model experiments on a combination of geometric [138], [140], [141] and monocular depth estimation loss [142], [143]. The loss can further be instigated using depth regression loss from the estimator of indoor scene variability $\mathcal{L}_{depth}$, multiscale gradient loss for the learnable scaling factor needed to design the monocular depth estimators $\mathcal{L}_{grad}$, the multiscale average loss to determine the object scalability $\mathcal{L}_{normals}$, and multi-view depth regression loss to determine the depth-based analysis of multiple regres-

sors used to mask and reconstruction task $\mathcal{L}_{mv}$ to construct total loss. A $\mathcal{L}$ can additionally add these losses in proportion to calculate the total loss.

with $\alpha_{grad} = 1.0 = \alpha_{normals} = 1.0$, and $\alpha_{mv} = 0.2$,

$$\mathcal{L} = \mathcal{L}_{depth} + \alpha_{grad}\mathcal{L}_{grad} + \alpha_{normals}\mathcal{L}_{normals} + \alpha_{mv}\mathcal{L}_{mv}. \quad (2)$$

3) 3D RECONSTRUCTION FROM BOROBODUR RELIEFS

The Borobudur temple is a UNESCO World Heritage Site well known as the largest Buddha Temple in the world [144]. It is situated in Indonesia and holds almost 2672 bas-relief panels, 1460 narrative panels, and 1212 decorative panels containing five sections, each having separate Buddhist stories. Due to the transient behaviour of the temple engravings and architectural richness, some parts of the well-known architecture are already ruined and need some reinstalled for safety concerns. Only four panels, consisting of 160 reliefs (Figure 7) at the southeast corner, are now visible to the visitors.

With the efficient DL architectures nowadays, the paper proposes a DL-based 3D reconstruction method by determining the Soft Edges of the pictures as edge guidance information. We divide the proposed method into specific steps for a better explanation.

- The first stage of the network encoder part takes an input of a single monocular photo along with the guidance of the prior edge detected image. The encoder model includes a Densenet [26] architecture to extract feature details from the input H. The densenet network comprises Batch Normalization followed by Rectified Linear Unit Activation function. The pipeline uses this architecture to extract 1/8 of the authentic image quality output. Furthermore, an Atrous spatial pyramid pooling [27] layer is introduced to extract contextual knowledge.

- The initial aim of the decoder phase is to restore the resolutions extracted from the feature map to the original with a factor of 2. An edge guidance layer with the factor size is added with the output from previous layers to locate important geometrical features for feature restoration. These geometrical features result from extraction from 3D edge areas and a higher opacity edge map to enhance 3D edge points. This point represents a restoration feature on the 2D image plane, thus added with the particular factor size in consequent stages. The edge guidance layer is motivated by a Local Planer Guidance (LPG) layer that captures more details during upsampling stage [145]. The restoration of features in LPG works well in the case of indoor and outdoor but drastically fails when applied to substitute datasets with higher curvatures and architectural patterns. To address this problem, an output with spatial dimension H/K passes through the subsequent layers to estimate a 4D coefficient that can fit a $K \times K$ patch to concatenate with the final prediction on the last convolutional layer.

$$D(y, y') = \frac{1}{n} \sum_i d_i^2 - \frac{\lambda}{n^2} (\sum_i d_i)^2, \quad (3)$$

$$d_i = \log y_i - \log y'_i. \quad (4)$$

This paper uses a scale-invariant loss to predict the depth map y and y' . The loss is predicted as the summation of minimal distances d_i with an average function added with squared invariance averaging from a squared number of samples n . The distance d_i is predicted log of the predicted value. The SPBR is a fast and 3D-based visualization method that works with a point cloud subgroup followed by a projection of 3D points for averaging the 3D subgroup module to create a final transparent image. It also ensures Opacity-based Edge highlighting by changing the curvature and linearity.

4) RESTORATION OF PARAMETRIC GEOMETRIC PATTERN BY ZERNIKE MOMENTS

Parametric geometrical patterns such as ancient Islamic architectures are an extraordinary treasury for mathematical algorithms. None of the objects ever last, and the historical architectures are the significant ones in the way of *mahaprasthan* (significant departure). Though multiple attempts have been made to restore and preserve the most artistic buildings in India, some aspects are still not visited in the case of various parametric geometric patterns. In this section, we will focus the study on the famous Sheikh Lotfollah and Naseredin Mirzan Mansion and attempt to restore some of the significant parts using image processing and ML-based approaches [31]. Figure 8 mentions a simple workflow engaging the architecture mentioned in [31]. There are two main approaches to pattern matching and spatial feature learning for pattern recognition. The workflow can be divided into four simpler parts, such as:

- Input stage takes input a good quality parametric Islamic image with distinct geometrical features.
- Image processing includes various noise cancellation and other related methods that contribute to the maximum feature extraction by cleaning and enhancing. The first process in this stage consists of grayscale conversion followed by a noise reduction algorithm. The preprocessed image is then passed to the perspective-based feature elimination converting all the polar coordinates to cartesian. The resultant images are given to two structures to recognize each element and detect the boundary of deterioration.
- To recognize each element in the subsequent preprocessing phase, the images are converted into a binary image to find the contours and convex hull. Deterioration boundary detection is similar to the texture segmentation problem from an entropy filter.
- The pipeline extracts Hu moments from the contour maps in the individual element recognition. These moments are then passed into a Kmeans clustering algorithm to remove the clusters present in the feature and fill it in the next phase. The next stage of the pipeline proposes Zernike Mahotas, which is an orthogonal basis set (a set of functions for which pair integrals are zero). It is mainly a weighted average of the image pixel intensities.

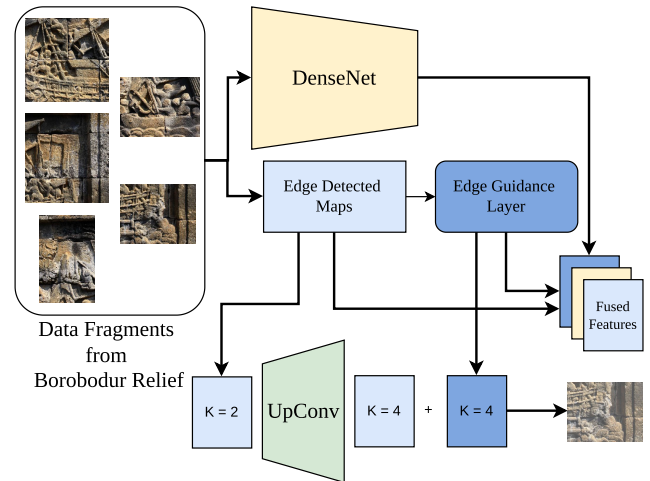


FIGURE 7. We discuss the method of reconstructing Borobudur relief using 2D convolution layer mapping Autoencoder. For the process evaluation, [144] considers fragments from the data to collect them through the interval cropping technique. The features from the data are extracted using DenseNet architecture in resemblance to an edge detection fragment sent through the edge guidance layer. These features are then tied, and the output image can be revamped in the last layer. The image follows layerwise propagation through up convolution operation, thus exhibiting a significant element perspective from the input fragment, which is the factor of an Autoencoder.

- The paper predicts the visual features by creating a time series concatenation. This time series data is passed into a DL framework for further reconstruction setting Mean Squared Error as evaluation metrics. The predicted restoration can further overlap with the deterioration region to visualize an extended pattern by calculating the overlap.

5) VIRTUAL RESTORATION OF THE FORBIDDEN CITY USING DEEP LEARNING

Colored paintings of ancient Chinese buildings have deteriorated in many ways, such as blurring, paint loss, and color distortion. There can be multiple defects on a particular tile, such as problematic hue and saturation, green becoming deeper, making golden edges hazy or blurry. Dragon patterns somewhat look a bit unclear or dim rather than glaring. The mural contains multiple cracks keeping a regular hue and low saturation based on the haziness of the tile. While various approaches addressed similar problems in several ways, the proposed method in [146] has created three different sections for the virtual restoration task, i.e., background, golden edges, and the pattern of dragon designs. The paper advises applying a benchmark for content segmentation using UNET [147] MobileNet [148], followed by a classical image processing approach for golden edge detection. Further improvements of dragons are rebuilt using GauGAN [28], an updated version of Pix2Pix GAN [149] architecture to generate realistic dragon patterns in the newly reconstructed area. Furthermore, the three results have been superimposed to fulfill the repair.

Most defects are beyond physical repair keeping the quality measurement intact and exact to the point base reversal technology. This technology defines the similarity between

old and newly made architecture, keeping minimum defects and the slightest difference measures. Extracting computational tools for the related work encourages the engagement of image processing, computer vision, and DL-based approaches to reconstruct the Region Of Interest (ROI). The measurement of contextual lectures regarding the pipeline can be highlighted as the critical conditions below.

- The first step of the pipeline introduces UNET based semantic segmentation method to obtain a color segmentation map on the tested image. The paper updates the standard UNET architecture with a MobileNet feature extractor resulting in a faster and more efficient segmentation result. Subsequently, two segmentation result is extracted to map the color of the second segmentation.

- This corresponding step offers a perspective transformation to set the particular segmentation map in a lovely position based on the black pixel background.

- Image processing approaches motivate enhancing edge detection criteria from the particularly transformed segmentation from two images. These two images are concatenated to create Complete Edge (CE) information. This information holds particular features and is fed into morphological transformations such as erosion and dilation to generate a Final Edge (FE) image. Using Erosion and Dilation finds the local minimum and maximum value that makes the image thinner making the boundary crisper and more extensive, making the edge more prominent.

- The application of the previous page extracts FE images as extensive background removal and edge-detected architecture. The pipeline further superimposes the image to generate the dragon picture based on human guidance using GauGAN. We can add this result with the superimposed image that restores the actual appearance of dragon figures, referring to a human-based architecture mapping, the beauty of GAN, and promotes computation technique for the restoration task.

6) GAN-BASED 3D RECONSTRUCTION

Generative Adversarial Network (GAN) [43] represents a generator-based fake image generation followed by a discriminator-ruled checking mechanism to generate realistic images. The generator captures the feature from the authentic images. It generates counterfeit images based on the components, while the discriminator tries to quantify the distinction between the fake (generated image) and the actual image. As there is an infinite number of elements available in nature with different physical characteristics, GAN possesses one of the largest experimental motivations to work with larger applications exhibiting various structures that automatically follow variations in Deep Generative models for different scenarios. Human faces were the first of its application using Deep Convolution GAN [24] with a convolution neural network-based generator and discriminator for better visualization in generated images. Pix2Pix [149] comes with a solution of image to image translations problems. This type of architecture introduces a new label map-based reconstruction using colorization and edge mapping along with learning

a loss function for different maps. Different modalities can add more scrutinized details in the generation of high-quality images. This idea is turned up with a new stacked architecture StackGAN [150], using multiple stage-based improvements proposing a novel augmentation technique to smoothen the images to remove sharper edge generation in fake images. In case of heritage restoration, multiple works have been carried out using various GAN architectures, improving the quality and effects by engaging super-resolution [151], enhanced super-resolution [152] with a better architecture for heritage-based reconstruction.

The *Borobudur temple of Indonesia*, which possesses a significant Buddhist relief collection, has some portions of reliefs hidden by stone walls. Only grayscale monocular photographs are now available for those hidden portions. A deep-based 3D reconstructed model from such a portion's grayscale photograph by Pan et al. in [153]. This method which can rebuild the invisible pieces into point clouds, was bettered by them by 5% in [30] and further by 2% as stated in [144] to achieve 97% accuracy. For the visible portions, the point cloud was obtained by PH scanning. Hence, the point clouds are accepted by the combination of DL and PH by a proposed method called stochastic point-based rendering to achieve a 3D transparent visualization. This visualization can portray the temple's hidden and visible portions.

Geometric patterns are commonly visible in historical Islamic architecture. Restoration of missing or damaged portions containing parametric geometric patterns is challenging. Hajebi and Hajebi [31] proposed a restoration method with an accuracy of 92%. The boundaries of the damaged portions are detected by texture segmentation using entropy filters. The extracted time series features to train a neural network to predict and reconstruct the missing patterns. This pattern restoration method has been tested on the dome of *Sheikh Lotfollah mosque, Iran* and *Naseredin Mirza mansion*.

Colored paintings on the surface of buildings suffer deterioration due to changing weather conditions. Zou et al. [146] proposed a deep-based method for virtually restoring the weathered beams of *Forbidden City*, the largest existing wooden structure complex in China. The image of the painting is divided into the background, golden edges, and dragon patterns, and then the parts are separately restored. Semantic segmentation with U-Net MobileNet is used for the background; traditional Canny Edge Detection is used for obtaining the golden edges. Hence, the three portions generated individually are superimposed to receive the complete regenerated image.

Kenya et al. presented a dense 3D reconstruction in [32] using an image-based pipeline in combination with a neural network. WireNetV2, a DL network, has been developed, which produces better results than other semantic segmentation approaches for masking images of a complex grid structure. This approach has been applied to the ideas of *Shukhov Radio Tower* of Moscow obtained by unmanned aerial systems(UAS) for its 3D reconstruction. *Generative Adversarial Networks(GANs)* [43] can be used in image restoration work.

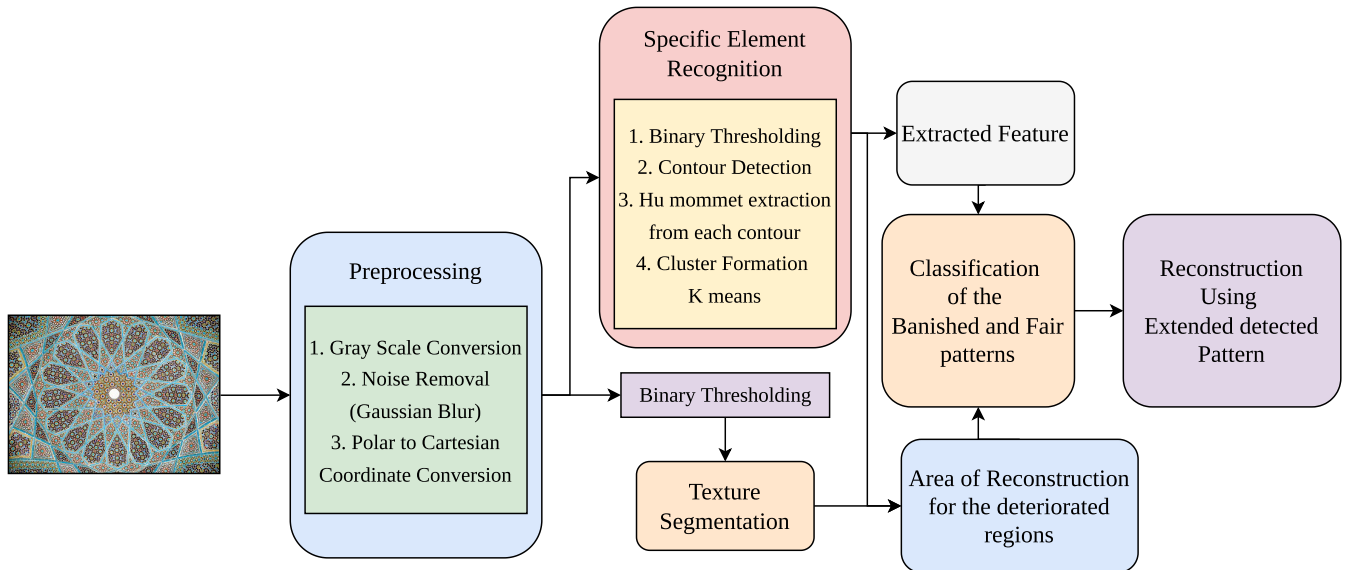


FIGURE 8. We elaborate on the process of Islamic pattern reconstruction using multi-label image processing and clustering-based segmentation headed by classification. The process states a classification task identifying defective and non-defective tiles followed by the borderline lengthening of entity reconstruction.

Kim et al. presented restoring old documents using Super Resolution GAN in [154]. The image generation network consists of a star network. A deep neural network and a hostile generative network are proposed for reconstructing images based on surrounding images.

Deep convolutional networks (CNNs) extensively extracted descriptive features from input images. In the recent past, the inverse problem of generating images has been dealt in [155], [156], and [157]. Attribute-conditioned deep variational autoencoders have been used in [158] by Yan et al. for the generation of images. A coarse-to-fine convolutional decoder generates local pixels based on entangled representation generated through multilayer perceptrons. But these images generated by VAEs are overly smooth. This has been addressed in [40] by combining a VAE and a Deep Convolutional GAN (DCGAN) to generate sharper and more realistic images.

It is difficult to process ancient documents and artifacts due to their natural degradation, fading, tearing etc. Images of palimpsests (overwritten parchments) and petroglyphs (rock engravings) very commonly contain partially erased or occluded portions. In general segmentation and binarization are applied on images for detecting and recovering characters or symbols against the background from visible portions of the data. However, the invisible portions cannot be handled by these methods. Melnik et al. [159] focused on recovering data from visible and invisible portions of corrupted images. A deep network consisting of a classifier and a conditional adversarial network (cGAN) has been introduced which can segment visible as well as missing or corrupted regions of an image.

Protection of ancient murals is essential. Digital inpainting or deep-based inpainting methods can repair damaged portions of ancient artifacts. Cao et al. in [160] proposed

a consistency-enhanced GAN for repairing missing areas in damaged murals. Image features are extracted from the fully convolutional network (FCN) and then applying deconvolution; the features are mapped to the original image size, and the output obtained is the repaired image.

IV. INPAINTING: A TRADITIONAL AND DEEP-BASED APPROACH

A. DIGITAL INPAINTING

Image inpainting [33], [161] is an image completion technique used to restore damaged images, to remove or restore some portions of any image with no visually perceptible difference.

Image inpainting finds crucial applications in the digital preservation and restoration of cultural heritage artifacts like murals, old paintings, manuscripts, documents, and photographs. These artifacts experience contusion and degradation over time or may get damaged due to scratches and folds. In such cases, inpainting is used to reverse the effect of time and artificially restore their undamaged version by recovering the damages and filling up holes or missing parts.

Technically speaking, inpainting is estimating missing pixel values in the damaged using the pixel information from the surrounding areas. Some approaches consider the immediate surroundings for this prediction, while others look for portions in the image that have the highest similarity to the neighborhood of the damaged section. Mathematically, inpainting is an ill-posed inverse problem since there can be many possible solutions, i.e., reconstructions corresponding to each image. Algorithms differ from each other depending on how realistically they can fill up the missing parts.

Inpainting methods can be broadly categorized into two classes: 1. Traditional approaches 2. DL-induced approaches.

1) **Traditional approaches:** Traditional inpainting approaches are suitable for filling small regions in images, like cracks and small holes. These techniques do not require training over the dataset but are unsuitable for images with large missing portions. The initial application of digital inpainting was made for image sequences or films [162], where missing regions in a frame are interpolated using information from adjacent frames. Later texture-based inpainting was proposed where a given area is filled with user-defined texture [163]. The first algorithm for image inpainting without any user intervention was proposed by Bertalmio et al. [33]. Traditional inpainting techniques can be broadly organized into five classes.

- Diffusion-based algorithm proposed by Bertalmio et al. [33] used discretized version of the anisotropic diffusion equation to fill up structured regions having different backgrounds. This approach attempts to imitate the manual inpainting methods through an iterative process. The central concept here is to smoothly propagate information from the neighborhood into the whole region along the direction of 'isophotes,' i.e., The direction of minimal intensity changes and is obtained by computing the perpendicular gradient vector at every pixel. Another variation was proposed [164], where the Navier-Stokes equations of fluid dynamics govern the information propagation along isophotes, and the flow of incompressible fluid models the image intensity. Chan [165] proposed a total variation (TV) method for image inpainting, where a total variational norm-based energy function is minimized. This method is further improved [166] by allowing curvature-dependent diffusion (CDD) for clean and noisy non-textured images. Here, nonlinear second and third-order PDE-based methods for inpainting to overcome the limitations of linear interpolations. These PDE-based methods are suitable for inpainting small regions and thin cracks.

- Texture synthesis-based methods, also called patch-based synthesis, pick up sample textures from clear sections of the image and replicate them in the damaged region, maintaining the color relation between the newly constructed region and its neighbourhood. Ružić and Pižurica [167] proposed a context-aware texture-based inpainting method. Here the image is divided into blocks of variable size based on their context, and the approach searches non-local image regions for well-matched patches. The process uses Markov Random Field (MRF) prior, i.e., a priori knowledge about the consistency of neighbouring image patches modelled using MRF. Kawai et al. [168] proposed an inpainting method based on *diminished reality*, a background geometry-dependent gap-filling method. This approach approximates the background geometry of any object by combining the local planes and incorpo-

rating perspective correction of textures. Some inpainting approaches use a low-rank prior [169], i.e., any raw image converted into a low-rank matrix owing to the non-local self-similarity. Thus the inpainting method boils down to appropriate singular value truncation. Lu et al. [170] use low-rank approximation on horizontal and vertical image gradients. Zeng et al. [171] proposed an inpainting method that uses a saliency map and color information to locate the best matching patch in the image.

- Exemplar-based methods combine diffusion-based and texture-based inpainting advantages. The approach was first proposed by Criminisi et al. [172]. The algorithm first divides the image into two segments: the target region, which is to be inpainted, and the source region, which is the rest of the image. Samples from the source region are selected and used to fill the target region. The filling-up process starts at the contour of the target region and moves inwards to the target. For each pixel on the contour, a fixed-size patch is constructed, having the pixel at its center. Then for each patch, a priority is calculated based on a *confidence value*, which measures the ratio of available pixels to total pixels in a patch, and a *data value*, which measures the strength of isophotes in that patch. Having determined the priorities, the highest priority patch is filled up first, using a patch from the source region with the highest similarity with the target patch. In [172], the authors have used the sum of square difference. After filling up every patch, the confidence values are recalculated, and the iteration proceeds. The author has demonstrated that filling orders are crucial for faithful inpainting. The exemplar-based methods cannot properly reconstruct multiple textures and geometric line segments. [173].

Different modifications of this basic exemplar-based approach have been developed by adopting different methods for calculating the data values, confidence values, and patch priority [174], [175], [176] and employing different search strategies [177], [178]. Hung et al. [179] proposed an improved method by first applying mean-shift segmentation to extract the contours, then repairing the broken contours by using Bezier curves, and finally applying the traditional exemplar-based inpainting. Another approach [180] uses weighted aggregation of multiple non-local neighborhoods that exploits the redundant information present in any image to generate better quality results. Fan et al. [181] used an adaptive patch searching method by changing patch sizes depending on patch sparsity and picking up the most similar rotation invariant patch matching the target patch. Perceptually aware inpainting [182] is another modification of exemplar-based inpainting that uses perceptual-fidelity aware mean squared error (PAMSE) [183] to measure

patch similarity. PAMSE is a Gaussian-smoothed MSE, i.e., a weighted sum of MSE, gradient, and Laplacian of MSE. The proposed approach reconstructs a large region having multiple textures and structures.

- Hybrid methods amalgamate PDE-based inpainting and texture-based inpainting. The input image is first decomposed into structure image, and texture image [184]. Then the structure part is inpainted using diffusion-based inpainting, and the texture image is processed with a texture synthesis method. Finally, the two resultant images are combined to get the final inpainted image [185]. In [186], the structure/texture decomposition of the image is done by a nonlinear diffusion filter of Perona–Malik type [187]. The geometry section is inpainted using Ginzburg–Landau algorithm [188], and then a coherence-enhancing anisotropic diffusion filter is applied to remove unwanted kinks and sharp corners generated from inpainting. The resultant image is then segmented, and texture synthesis is completed. Wu and Ruan [189] used a total variational method with an L1 norm regularizer to decompose the image. Then the structure part is inpainted with bi-directional diffusion PDE, and the texture part is synthesized using exemplar-based inpainting. In [190], the authors apply a finite element-based Bi-dimensional Empirical Mode.

Decomposition (FE-BEMD) to decompose each channel of the image into a set of Intrinsic Mode Functions (IMFs) and residue. Then exemplar-based inpainting is applied individually to all the IMFs and residue, and finally, the results are added together. Cai and Tim [191] proposed a context-driven search window method, where the window size depends on how texture-intensive or structure-intensive the target block is. In general, hybrid methods have high computational complexity.

- 2) **Deep-based Inpainting:** DL-induced inpainting methods [192], [193], [194] are popular with the advent of various promising DL algorithms and architectures in computer vision and image processing. These approaches outperform traditional inpainting methods when the damage in the images is enormous. Here DL-based approaches are categorized based on the network architectures used.

- CNN-based approaches: Pathak et al. [195] proposed a context encoder, which is an encoder-decoder type architecture for reconstructing the missing region of any image based on its learned features. The encoder is based on the AlexNet [60] architecture having channel-wise fully connected layers. The authors used a composite loss function comprised of l_2 norm-based reconstruction loss and GAN-based adversarial loss. Here, the encoder encodes the image and the missing region into a suitable latent space, and the decoder attempts to reconstruct it using upsampling. Thus the

features learned from the image segment help reconstruct the masked region. In [196], the authors have proposed a blind inpainting convolutional neural network (BICNN), which can reconstruct the damaged image without prior knowledge about the location inpainted. The network uses convolution layers with zero padding and ReLU activation with no pooling or fully-connected layers. The network is trained on preprocessed ImageNet dataset by minimizing MSE loss. Sasaki et al. [197] proposed a CNN-based network for automatically identifying and completing the gaps without any user intervention for line drawing images. The network consists of zero-padded convolution layers with ReLU and upsampling layers.

Input images are grayscale line drawings of any size, and generated output is the same size as the input. Yang et al. [198] proposed an encoder-decoder CNN network using a loss function containing three terms: a holistic content term, a local texture term, and a TV-loss term. There is an encoder-decoder type content prediction network, which employs a combination of l_2 loss and adversarial loss to learn to predict the content of the removed region of the input image. A pretrained VGG-19 is used as a texture network, and the outputs of *relu3_1* and *relu4_1* layers are combined with learning local textures. The TV-loss term invokes smoothness in the reconstruction.

Moreover, the authors have proposed a multi-scale neural network for patch synthesis in high-resolution images. A U-Net-based architecture with a shift operator is proposed in [199]. The network is named Shift-Net. Its shift operator in the encoder helps to reconstruct the missing parts using the features learned from the known region. For training, the authors employed a combination of reconstruction loss, adversarial loss, and guidance loss, which the shift operator incurs. The guidance loss is the squared norm of the difference between the ground truth image's encoded feature and the recovered image's decoder feature. The corresponding encoder and decoder layers are connected via the shift operation. Artist net [200] is also an encoder-decoder type model that separately extracts and compares the style features and content features of the ground truth image and the synthesized image. It used an intricate loss function to ensure the reconstructed image's content and style consistency. The loss function contains two terms: a reconstruction loss comprising of l_1 loss between reconstructed and ground-truth image and perceptual loss [201]; a latent loss containing content loss and style loss in latent dimension and finally, an adversarial loss computed using a PatchGAN discriminator.

One drawback of the CNN-based approaches is that the convolution operator works on both the actual image pixels and the pixel values of the masked holes. This

causes spurious artifacts in the reconstructed image. Segmentation-aware convolutions through partial convolution layers [202] are applied to salvage the problem. Partial convolution operates only on the original image pixels in each layer. Another attempt by Yu et al. [81] was made utilizing gated convolution having learnable soft masks.

- Yeh et al. [40] proposed an improvement over the Context Encoder [195] using deep generative models. A GAN is trained on a set of uncorrupted images to obtain faithful and realistic reconstructions. Now, given any masked image, it's encoded in some latent vector (z_m) by the discriminator, and then the latent vector z_i , "closest" to z_m , is searched in the latent space. The latent vector z_i corresponds to an uncorrupted image in the training dataset closest to the masked image. Finally, the generator uses the latent vector z_i to reconstruct an image that corresponds to the reconstructed image. To measure the closeness of the latent vectors, the authors used a loss function comprising of a *content loss* and *prior loss*. The content loss is the weighted $l1$ distance between the corrupted image and generator output with z_i ; the prior loss, which ensures that the generated image lies in the same manifold, is the GAN loss used during discriminator training. A similar procedure has been used with an improved version of Wasserstein GAN [203] by Vitoria et al. [204]. The modifications are threefold: 1. residual blocks are incorporated in both generator and discriminator of the WGAN 2. the fully connected layers are eliminated 3. the traditional batch-normalization layers are replaced with layer normalization layers. Also, the authors added a new loss term, namely *gradient loss*, which is a weighted $l1$ distance between the gradient of the masked image and the gradient of the recovered image. Jboor et al. [205] followed this same basic principle for inpainting particularly different types of visual cultural heritage datasets to inpaint any given paintings or portraits etc. The authors considered the WikiArt [206] dataset, the MET dataset [207], and the Rijksmuseum dataset [23]. They have employed VGG16 feature vectors, SIFT features [208], and SURF features [209] for clustering different cultural images. Then separately trained DCGANs individually for each cluster. When any incomplete image is given, the algorithm first detects the cluster where the input image belongs, and then the corresponding GAN is used to reconstruct the damaged image.

The aforementioned GAN-based methods reconstruct a single image as the solution of the masked image. However, that generated image may not be a convincing and realistic one. As an alternative, an architecture PiiGAN (Pluralistic Image Inpainting GAN) [210] was proposed, which can generate several reconstructions consistent with the source image. The network uses a style extractor and feeds the masked image and the

style vector to the generator's input. It also employs two discriminators for patch-based and global prediction. The effectivity of the architecture is studied on CelebA, PlantVillage, and MauFlex datasets.

- 3D inpainting remains a growing field of interest with the increasing popularity of 3D digitization of cultural heritage sites. 3D data acquisition, 3D data processing, and 3D reconstruction from point cloud data have been greatly emphasized by researchers worldwide. A problem encountered during 3D digitization is gaps and holes in the 3D point cloud data due to the unavailability of sensor data or naturally occurring damages. This, in turn, creates visually unpleasing and incomplete 3D reconstruction. To salvage this problem, 3D inpainting techniques are used, which can fill up the holes in broken and damaged models [34], [211], [212] and create a realistic completion.

- Image inpainting can be used for image forgery by eliminating particular objects in any digital image without perceptible changes. It necessitates developing methods that can localize inpainted regions in any image. Several approaches [213], [214], [215] detect inpainted regions for exemplar-based inpainting. Liang et al. [215] proposed a forgery detection for detecting exemplar-based inpainting in any image. This approach combines central pixel mapping (CPM), greatest zero-connectivity component labeling (GZCL), and fragment splicing detection (FSD).

Zhang et al. [216] propose a forgery detection approach that is robust to processing like image compression, noise addition, and many more. Although in-painted segments are visually unrecognizable, the correlation discrete cosine transform (DCT) coefficients of the in-painted patch and neighboring regions are disturbed. This correlation reduction is especially noted in natural images and can be easily detected by exploiting the joint probability density matrix of the image pixels.

Detection of diffusion-based inpainting is much more complicated than identifying exemplar-based inpainting since diffusion-based inpainting covers much smaller and narrower regions than exemplar-based. Li et al. [217] attempted to detect diffusion-based inpainted regions in images. The key idea of the algorithm is that the image Laplacian along the direction perpendicular to the gradient behaves differently in the inpainted and intemperate regions. The authors propose a set of features based on the changes' intra-channel and inter-channel local variations.

There are several DL-based inpainting detection approaches. In [218], a CNN-based network is proposed that identifies texture-based inpainting. The network has an encoder-decoder structure and predicts the inpainting probability for each pixel in an image. The network is trained with MIT place dataset [219] with cross-entropy loss, and the testing is done with UCID dataset [220].

There are numerous works regarding digital inpainting, all of which can't be covered in this survey. Some other surveys [192], [193], [221], [222] that have focused only on digital inpainting and hence covered many more research papers, which the reader can consult for further study.

V. 3D SEMANTIC SEGMENTATION: MULTI-VIEW AND POINT CLOUD-BASED APPROACHES

Segmentation on 2-dimensional images motivates extracting meaningful artifacts from the picture by calculating probable background structures based on DL and image processing approaches [147], [223]. On the other hand, multi-view geometry better use multiple camera scenes to observe a 3-dimensional spatial relationship of calibrated images engaging numerous viewpoints. Connecting segmentation and multi-view geometry can be a well-defined work to support concrete architecture-based point cloud detection expressing the background and subjective object from the scene. This part of the survey discusses some crucial 3D segmentation approaches with multi-camera segmentation for motion-sensing images and videos without creating a dense 3D reconstruction algorithm. It further explains the 3D segmentation of meaningful objects rather than 2D, imposing more significant results.

Multi-view object segmentation is an advantageous task to be addressed using 3D spatial segmentation, engaging several viewpoints rather than limited resource monocular approaches [46]. The method uses variational inference to improve the applicability of the sparse modelling to combine GMM (Gaussian Mixture models). Next, they propose a depth estimation, minimizing the energy mapping using colour-based segmentation to extract multiple Regions of Interest between foreground and background. Time series estimation can be a vital scenario considering further improvement of object tracking from motion pictures. Thus, numerous viewpoints are engaged to make a better real-time segmentation irrespective of shadow and imperfect connection between background and foreground using mesh tracking and 3D reconstruction [224].

On the other hand, point cloud based approaches have been extensively discussed with an attention to ML and DL approaches. DeepFit [225] demonstrates a technique for fitting point clouds to 3D surfaces using the DeepFit neural network architecture. It performs better than currently used techniques and can handle noisy and imperfect data. [226] discusses a generative model for point clouds that captures long-range interdependence through self-attention mechanisms. It can produce 3D models that are lifelike and have precise details.

A. INTERACTIVE GRAPHICS FOR HERITAGE DATA RECONSTRUCTION

Point clouds are sample points that depict an object's outer region. Several interactive computer graphics procedures can transform these points into vectorized 3D models. Construc-

tive Solid Geometry (CSG) [227], which dissects the point cloud model into its vectorial components and textures them accordingly, is one such solution. Using Boolean operations like union, intersection, and difference, simple geometric forms are put together to generate a 3D representation in this procedure. The final model accurately captures the raw point cloud data.

The effort of CSG in 3D model vectorization from point cloud data with interactive computer approaches covers a much more prominent aspect in detail. [228] suggests using deep learning to rebuild CSG objects' volumes from disorganized point clouds. The proposed framework, which consists of an encoder-decoder network and a CSG operator network, makes it possible to rebuild CSG objects more precisely and effectively than with conventional techniques. Advertently, [229] proposes an iterative optimization algorithm-based surface reconstruction technique for CAD models with stored CSG information. In order to create precise and supple CAD models, the method extracts the boundary surfaces of CSG objects and optimises them using a boundary element method. The reconstruction of CAD models from point cloud data is proposed by [230]. The approach extracts the geometric structures of CAD models using CSG operations and a voxel-based representation of point clouds. These geometric structures are then improved using a local optimization strategy. [231] proposes an updated surface reconstruction strategy is an important aspect for old building reconstruction thus empowering digital heritage archive. This method use voxel based representation to reconstruct 3D model using CSG from point cloud data.

• CSG for heritage data

CSG for heritage reconstruction can have a better impact for 3D model reconstruction from point cloud data. The procedure of terrestrial laser scanning helps to capture 3D data. Thus, [232] use constructive solid geometry (CSG) operations to combine the individual scans into a single model. CSG is a method for creating three-dimensional (3D) objects out of basic geometric shapes like cubes, cylinders, and various object shapes. Each node in a tree of CSG model structure represents a basic shape, and the internal nodes stand in for the Boolean operations that combine the basic shapes. In extent, point cloud data from multiple sources can result a more accurate 3D model [233]. With more complex details, LiDAR point cloud as input data [234] results a higher resolution detail present from the picture of the source representation. [235] compares the precision and depth of laser scanning data for cultural heritage objects with 3D photorealistic rendering models. The paper examines the 3D models' quality and potential for heritage reconstruction employing CSG techniques. The geometric feature representation from objects in [236] presents a CSG-based modelling strategy for historical architectural items. The method builds precise 3D models utilizing point cloud data and then employ CSG techniques to describe the geometric properties of the object. For purposes of reconstructing and documenting cultural heritage, the suggested method is appropriate.

CSG is now a crucial component of processes for both BIM and HBIM [237], [238], or historic building information modelling. The architectural, engineering, and construction (AEC) business uses BIM, which is a graphical portrayal of a building's structural and functional details. The requirements of historic buildings [239], which are distinguished by intricate geometries [240], distinctive materials, and historical significance, are addressed by HBIM, a BIM paradigm. CSG is a mechanism that may be used to model complex architectural components in BIM and HBIM, including windows, doors, staircases, and facades. Such elements can be easily manipulated and subjected to precise measurements and analysis by being represented as CSG objects. Furthermore, CSG may be used to characterize intricate connections between various building components, which can be utilized to automate a number of processes like collision identification, cost estimation, and energy performance monitoring.

In conclusion, CSG is a potent modelling method that can serve as the foundation for BIM [241] and HBIM workflows. Making use of CSG's advantages makes it feasible to build precise models of intricate architectural components that can be utilized for design, research, and documentation projects, among other things.

Apart from CSG, multiple interactive graphics approach is proposed to express the keen interest of the community nowadays. A web-based interactive visualization system for substantial urban reconstructions is presented in [242]. The system enables exploration of variously detailed 3D reconstructions of cities and structures. It offers real-time interaction with the models and has a progressive loading system to accommodate huge models. Taking a step back from web-based module, the presentation and exploration of cultural heritage places using dispersed, mixed, and augmented reality technology is covered in [243]. It provides an extensive review of these technologies and discusses some potential uses for them in the archeological domain. The article also includes a variety of examples demonstrating the usage of these technologies in realistic heritage contexts. [244] presents an interactive 3D modeling tool for the digitization of cultural heritage sites. The application provides plenty of instruments to create sophisticated 3D models from images and other data sources and is intended to be simple to use. The report includes numerous examples of the usage of the tool in legitimate heritage settings. In [245], a Neolithic jadeite axe is used as a case study for the use of VR to visualise a museum collection. The research outlines the process for creating a 3D model out of images and a laser scan and showing it in virtual reality. The essay also discusses the benefits of using virtual reality to display artefacts and collections of cultural heritage. [246] demonstrates an interactive 3D visualisation end product system for archaeological excavations that allows users to interactively explore and modify 3D reconstructions of excavation data. The paper additionally focuses on the challenges in preserving cultural heritage sites and the aspects wherein technology may enhance cultural heritage preservation and presentation.

B. AUTOMATIC SEGMENTATION OF HERITAGE DATA

In recent years, there has been considerable progress in developing automatic procedures to segment and classify point clouds or polygonal meshes. In this direction, Grilli and Remondino [47] developed a method to (i) document architecture-based historical information, (ii) distinguish between different construction techniques, and (iii) recognize existing evidence of restoration. They aimed to explore how ML approaches can be applied to cultural heritage.

The proposed method is a texture-based approach with supervised or unsupervised strategies to analyze the considered 3D structure. The method used Random Forest for supervised learning, and k-means clustering has been employed for the unsupervised segmentation. The performance of the model evaluated the overall accuracy. This methodology is effective for restoration when applied to four different heritage scenarios in Italy -

- *Pecile's wall of Villa Adriana in Tivoli* - Original and restored portions extracted from the digital wall.
- *Renaissance building in Bologna* - Different architectural elements were classified.
- *The Sarcophagus of the Spouses* - Identification of the surface anomalies like fractures and decays of this terracotta sculpture.
- *The Bartoccini's Tomb in Tarquinia* - Painted portions of the wall and the deteriorated portions of this 4th century tomb could be automatically identified.

C. PointNet

Convolutional architectures require regular input data formats such as 3D voxels or image grids to perform operations like kernel optimizations. Point clouds represented by a set of 3D points $\{P_i | i = 1, 2, \dots, n\}$ have an irregular format. Since they are an essential data type, many researchers transform point clouds into 3D voxel grids. This transformation renders an unnecessary increase in data volume. Qi et al. in [45] designed a novel neural network to handle this. This network, named *PointNet*, considered point clouds as input well, respecting the permutation invariance of the points. Each point has been represented by three coordinates (x, y, z) , and additional dimensions can be added from the computation of normals or other features. *PointNet* uses a single symmetric function, max pooling, which is quite effective yet simple. On learning a set of optimization functions, the network selects some points of interest in the point cloud and encodes the reason for such selection. The input is either sampled from a shape directly or a part is segmented from a 3D scene. The final fully connected layer aggregates into the global descriptor, the optimal learned values for obtaining the entire shape. The output for each n point and m semantic subcategory will be $m \times n$ score. It is convenient to apply rigid or affine transformations to the input format of this network as each point transforms independently. Also, an advantage of this network is that it learns summarizing of an input point cloud

by sparse points. Moreover, it is highly robust to small input perturbations or corruptions due to outliers and missing data. A thorough theoretical analysis and an empirical evaluation showed that the *PointsNets* exhibited at par or even better performances than the state of art.

D. RANDOM FOREST BASED POINT CLOUD EXTRACTION FOR HERITAGE DATA PRESERVATION

As 3D point cloud information become more readily available, there is a massive growth for reliable and efficient methods for obtaining valuable knowledge from these data sets. The usage of random forests [247] is a popular strategy that has shown promising outcomes in a variety of fields, including point cloud classification, segmentation, and feature extraction. The use of random forests for point cloud extraction is particularly pertinent in the field of heritage data preservation since it enables the identification and preservation of significant cultural heritage aspects. In this regard, a number of research articles have examined the use of random forests to the extraction of point clouds [45] from heritage data. In the sections that follow, we will evaluate a few of these works and their significance.

1) AUTOMATIC SEGMENTATION OF POINT CLOUD DATA

[248] presents a new method for segmentation of point cloud data information for cultural heritage sites. The research provides an automatic segmentation method for point clouds of cultural heritage sites based on random forests. The proposed approach takes use of a set of geometric, radiometric, and contextual data for each pixel, as well as a hierarchical segmentation scheme based on an adaptive octree structure. The findings indicate that the suggested strategy outperforms other approaches in terms of segmentation accuracy.

2) POINT CLOUD CLASSIFICATION BY MEANS OF RANDOM FOREST AND SVM

A comparison between random forests and support vector machines for the classification of point clouds of historical buildings is demonstrated in [249]. The technique divide points into numerous classes, including such roof, wall, and windows, using morphological and radiometric properties. The study demonstrates that in regards to classification precision and computational efficiency, random forests beat support vector machines.

3) ML BASED APPROACH ON EGYPTIAN BASALT QUARRIES

This study on ancient Egyptian basalt quarries in [250] suggests utilising LiDAR data to classify using a random forest technique. The authors developed a digital surface model (DSM) and a digital terrain model by first cleaning the LiDAR point clouds to reduce noise. Then they take information from the DSM and Digital Terrain Model (DTM), such as height above ground, slope, and curvature. The random forest classifier, which divides the point clouds into the three classes of quarry, outcrop, and soil, is trained using these attributes.

The correctness of the technique as a whole, as reported by users, as reported by producers, and as measured by the Kappa coefficient are some of the criteria used by the authors to assess its success. Results indicate that the suggested strategy outperformed more conventional approaches like visual interpretation, with an overall accuracy of 93.7%.

The proposed method is innovative because it classifies ancient Egyptian basalt quarries using machine learning methods. The work highlights the possibilities of analyzing and categorizing archaeological features using LiDAR data and machine learning techniques.

E. RadDGCNN: A RADIUS DISTANCE BASED POINT CLOUD ARCHITECTURE

Preservation of cultural heritage is often tricky due to shape complexity and curve repeatability. [42] presents a point cloud-based Dynamic Graph Convolutional Neural Network (DGCNN) that exploits radius distance as an objective function as the limited availability of datasets in this field makes this task impossible for DL-based architectures. Terrestrial Laser Scanning (TLS) and other photogrammetry approaches generate 3D and geometrical coordinates around the scene to document this issue. However, creating 3D models from point clouds is time-consuming and needs partial automation in the Scan-to-BIM process.

The idea is to develop a classifier that can categorize a point cloud based on the knowledge of the data with a different color that can be known as a particular type of semantic segmentation in action. This model is evaluated with two publicly available datasets: the Ducal Palace dataset in Urbino and Palazzo Ferretti in Ancona, Italy. As a DL model needs a tremendous amount of annotated data which is quite tricky and time-consuming in the case of point cloud segregation, the paper proposes a Synthetic Point cloud extraction to train DL models for semantic segmentation.

As we delve deep into the methodology, the workflow is created based on the documentation of the process from point cloud extraction using HBIM and the IFC method. Preprocessing is crucial to providing the best segmentation point cloud extraction for further processing in the RadDGCNN model. This part includes sampling three-dimensional $2 \times 2 \times 2$ blocks followed by a geometric feature removal in the XYZ axis. Several classes have been selected for further experiments, e.g., Moulding, Column, Pilaster, Floor, Vault, and many more. Extraction of the point cloud from the dataset is processed in two parts. Firstly, point cloud extraction from a 3D model is considered using Scan-to-BIM modeling. Secondly, point clouds are extracted from monocular architectural 3D elements collected from online open repositories.

The final part of the workflow comes with the architecture of the deep neural network, DGCNN [29]. It starts with a 3D channel node representation with random rotation. The further layer processes an Edge-based convolutional radius-based mapping (ECM) with Multilayer Perceptron kernel network 64 (Figure 9). The additional layers are connected

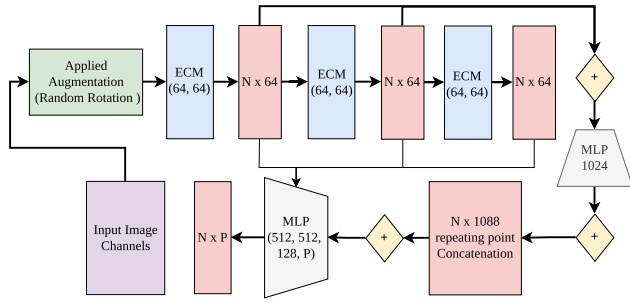


FIGURE 9. The architecture of Radius Distance mapping for semantic segmentation of architectural components from historical heritage sites. The network is similar to convolutions mapping but introduces an Edge guidance mapping vector instead of kernel mapping followed by algebraic matrix calculation.

with similar layers with updated channel information modeling until a skip connection step in the Fully connected part is applied using another Multilayer perceptron to keep the identity mapping. The Edge Convolutional layer is further considered a concatenation of Radius neighbors followed by MLP and max pooling operation.

F. 3D SEGMENTATION ON PETROGLYPHS

Petroglyphs or Rock engravings are essential historical artifacts to study and preserve human history and notions of analysis between the present with the past. Many are on the verge of collapse due to natural phenomena and human-made disasters. Several manual efforts have proven essential to document and preserve historical findings, along with the location tracing of thousands of rock engravings worldwide. However, this is a tremendous amount of time-consuming work considering the manual intervention during the data collection and location tracing of the correct engravings. Furthermore, manual photogrammetry-based methods engage 2D imaging losing the artifact's quality and depth mapping feature in the documentation stage.

Nowadays, 3D reconstruction and modeling are fashionable fields of interest. 3D mapping and segmentation of particular petroglyphs can have an advantage over manual effort. However, substantial 3D point cloud estimation and robust 3D reconstruction are highly time-consuming and computationally expensive. To address this issue, [49] proposes an image-to-space modifying segmentation approach that transfers the task to the adaptive learning-based interactive segmentation to create a gradient-preserving energy minimization. This technique can be considered a crucial preprocessing step before classifying the point clouds as foregrounds or backgrounds.

One of the first approaches in the related field is surface topography extraction [48]. This approach engages extraction of point cloud followed by smoothening filtering. Local deviations define the structural complexity of the surface points, defining the distance between any two selected points within the points and stretching it. Furthermore, 2D mapping from the 3D is generated using a nonlinear dimensionality

reduction technique. However, this method is highly computationally expensive, exhibiting satisfactory results.

[49] proposes an alternative way to compose the point clouds to a 2D depth map estimator based on projection direction. This depth map is generated on the projection from the heatmap-based intensity mapping on a support plane estimated from the surface area. The support plane is calculated using least squares distances to the particular region of the plane surface. The orthographic promontory of the 3D points on the support plane determines the 2D depth map of the 3D surface that can capture global geometry and local topography. Equation 5 presents a gaussian depth distribution filter $G(x, y)$ is imposed to smoothen the depth map by a convolution operation which is squared average shifted to the exponential mapping to represent the filtering.

$$G(x, y) = e^{\frac{-(x^2+y^2)}{\sigma^2}}. \quad (5)$$

The deviation map $\hat{D}(x, y)$ includes a standard divergence with the gaussian depth estimation filter tallied along with the window of W^2 to all possible locations equation 6. The mathematical formulation of the map instead explains the analysis between incremental source point i and j . In the context of the micro presence of architectural entities, this method works satisfactorily, representing the regional deviation depth map into all conceivable areas over the surface.

$$\hat{D}(x, y) = G * D = \sum_{i=-W/2}^{W/2} \sum_{j=-W/2}^{W/2} G(i, j)D(x + i, y + j). \quad (6)$$

Classification is essential to determining the probability of each generated point cloud. The paper used Random Forest [247] as a classifier taking the results of the Enhanced Depth Map directly as a feature vector on Camonica Valley of Brescia, which contains more than 140,000 rock engravings. The dataset was collected through Surface Laser Scanning (SLS) procedures, resulting in high-resolution scans of the petroglyphs.

G. LARGE-SCALE POINT CLOUD SEGMENTATION

Point Cloud Segmentation remains a classical problem in the context of 3D modeling and analysis of Historic Building Modeling data. Multiple works in the field engage HBIM algorithms to analyze the effect of Point Cloud extraction followed by classification segmentation in this field. However, neither is time nor information effective for large-scale point cloud analysis. Manual intervention is another example of a practical approach, but it takes time. While DL continuously replaces a plethora of classical algorithms with learning-based ones, cultural heritage is no different from welcoming automation using machine-level intelligence approaches. Moreover, these approaches are promising and time effective but need a massive dataset to train the model.

[50] aims to propose a large-scale 3D point cloud segmentation dataset names ArCH dataset² labeled with ten classes and divided into a train, test, and validation sets. The dataset consists of 17 annotated and ten nonannotated point clouds. Many UNESCO world heritage list sites are present in the dataset. Some of these are *Chapel of Strasbourg Cathedral*, *Courtroom of Valentino's castle*, *Sacro Monte of Varallo* and *Ghiffa*, *St. Pierre church*, *Porticoes of Bologna*.

The data acquisition process is the preliminary stage of modeling and innovation. Most scenes are taken using Terrestrial Laser scanning and Photogrammetry based approaches. FARO Focus 3D $\times$ 130 and 120, Riegl VZ-400 is used for TLS. Photogrammetry-based approaches are performed using Nikon D880E and Nikon D3100. Moreover, for Ultra Violet Scanning UAV platform was equipped with SONY Alice 5100L.

Data collected through TLS and Photogrammetry based scanning are mostly noise inclusive and thus need meaningful preprocessing in the later stages. For this purpose, ClodCompare is used with two different functions.

- Georeferencing the Cultural heritage site is quite important. However, this technique is costly and needs a small amount of human intervention to check the locations. The coordinates were truncated in every scene and shifted to the origin (0, 0, 0).

- Octree and space-based subsampling instead of random subsampling are used to tackle the vast data with limited test repeatability. 1% updates the test sets to generate the difference between the classes and labels.

Feature selection is the most critical approach after meaningful point cloud extraction using several preprocessing approaches. Such selections can now be effective for segmentation and classification. Here, the K Nearest Neighbor algorithm uses CloudCompare for the feature selection and segmentation task. Ten nearest neighbors have been used to classify the nature of the point cloud, followed by a feature extractor from Convolutional Neural Network layers. The point cloud structures are x, y, z as the coordinate in the 3D plane, r, g, b measures the colour values, corresponding *label*, number of pixels representing the coordinates N_x, N_y, N_z .

VI. CLASSIFICATION OF CULTURAL HERITAGE

Cultural heritage sites are divided into various types based on their structural complexities and the inception behind the visually present content. None of the classical algorithms can comprehend the classification as the structural complexities and features are carved deep into the architecture. The new era of ML and DL neither depends on classical approaches nor defends the complexities and richness present in the content. DL algorithms such as Convolution Neural Network [56] with efficient backpropagation [251] tend to solve cutting-edge problems. Such algorithms can extract architectural complexities, delve deep into the pixel values within the image, and back-propagate positive feature maps into the distribution of

weights for an update in the next stage for a specified target value. One can set the label of the specific class in which the image belongs into the very target value to decide how far the features are from the actual maps, thus segregating better positive components in the later steps.

A. INCEPTION PROJECT: APPLICATION OF CNN ON CULTURAL HERITAGE SITE

The extension of the tension of DL into Cultural heritage data has been propagated by [57]. This paper extends the analysis of DL into the inception of the "INCEPTION" European project to develop tools to prove the effectiveness of mathematical algorithms, segmentation, and HBIM techniques for cultural heritage assets. The project aims to reconstruct and innovate the 3D modeling of time-dynamic heritage data that will be accessible to all types of users. This approach should cross the social, cultural, and religious beliefs between collection and documentation by giving the end user a lightweight, well-defined visual experience. Traveling all the boundaries, the project aims to develop a search engine of historical modeling rather than a well defines pipeline programmer approach.

Transfer learning is a well-known method to extend the capability of a Convolution Neural Network (CNN) rather than training it entirely from scratch. This method introduces a pretrained network with a large dataset and fine-tunes it into a new dataset category. The process fixes the feature extractor part of the CNN, as it already knows how to extract meaningful features. The last layer of the CNN will be removed by the number of classes present in the new dataset, and extracted attributes will be propagated through the trainable dense layer to train the features for specific class label targets. The paper uses the INCEPTIONV3 model for the INCEPTION project with lower-level training by fixing the feature extractor that has been pretrained with the IMAGENET dataset. A set of Flickr datasets with 10,092 images has been used for transfer learning keeping 8000 for training, and 1000 for validation and testing.

Considering classification as an essential parameter for this project, the researchers increase their notion to a broader sense to some extent. [58] made public the dataset of 10000 images along with additional content of 1404 test data starting with other preprocessing methods presented in [60], and [59]. While AlexNet is the first and foremost stable Convolution Neural Network architecture, ResNet offers a better gradient flow using residual connections between the previous and next layer to pass the positive gradients, ensuring the result never gets worse through identity mapping.

[58] experiments with AlexNet, ResNet, and InceptionV3 on various image sizes and parameter settings based on image size and epoch. Such experimental settings have been conducted with full training by unfreezing all parameters and transfer learning by unfreezing only dense layers on ten specified class labels. Specific criteria of overfitting and underfitting thus need a good analysis to maintain the bias-variance tradeoff. This mechanism enforces Inception-ResNet-V2

²<http://archdataset.polito.it/>

with the fine-tuning setting on 128×128 image size for 77 epochs. Later, the researchers tested fifteen more categories on other networks to prove the efficiency of the model proposed in the paper. This paper presents a pleasant experimental understanding between full and fine-tuning with several deep-learning models of architectural differences. This process also offers a tradeoff between bias-variance, learning rate, and weight decay to present the effect of various hyperparameters on the classification task on cultural heritages under the INCEPTION project.

Classification of images is essential for the documentation and analysis of cultural heritages. In [57] and [58], DL-based techniques have been applied for classifying architectural heritage images through CNNs. Training the network on a newly created dataset has shown promising results, which is helpful for digital documentation related to architectural heritage.

B. POINT CLOUD CLASSIFICATION USING ML AND DL APPROACHES

2-dimensional geometry presents an abstract view of the object without affecting the depth and good quality assessment. On the other hand, annotation is a significant problem based on the classification approaches to 2D data. Multiple datasets can refer only to the facades of the building, such as eTRIMS [61], Facades dataset [64], CMP dataset [65]. The limitation of Semantic information segmentation with 3D modeling is one of the growing fields of interest that can comprehend better results with multiple post-learning mechanisms. Furthermore, a point cloud-based extraction using three-dimensional coordinates instead of two helps to characterize and describe a better view of the computational heritage architecture. However, such methods are computationally heavy and time-consuming.

The classification of point clouds has been presented in [47] with some of the popular ML and DL algorithms as a classifier. Heritage buildings mostly have different components, and perfect classification is a prerequisite to reconstructing those particular architectures. This paper aims to use this approach in two test environments: the temple of Neptune in Paestum and some of the well-known renaissance buildings in Bologna. Depending on the feature selection, such architectures have different classes and regions of interest. Thus, the selection and extraction of features have been standardized based on three methods:

- **Decentralised Coordinates:** Represents the local geometry near a center and decentralizes it to the near geometry for the extraction of x, y, z coordinates.
- **Radiometric value:** Extraction of 3 Dimensional RGB frames for the radiometric measure of the particular point followed by re-scaling.
- **Geometric Features:** We can capture multiple types of geometry from standard cultural heritage sites, such as Covariance features, height, depth-based characteristics, and standard and shade color-based features.

TABLE 2. The method sets 10 classes for the Greek temple of Neptune and 14 classes for the Renaissance building of Bologna.

Greek temple of Neptune	Renaissance building of Bologna
Grass	Road
Crepidoma	Pavement
Pavement	Facade
Shaft	Base
Echinus	Shaft
Abacus	Capital
Architrave	Arch
Frieze	Window / Door
Cornice	Vault
Tympanum	Moulding
	Curtain
	Vertical drainpipe
	Cornice
	Horizontal drainpipe

2 represents the class labels that made the classification task possible in the later stages. For supervised classification tasks based on the class label, Random Forest, One versus One classifier, Convolution neural network, Bi-LSTM is used. Moreover, the CNN architecture presents two types of varieties as 1D and 2D CNN. 1D CNN consists of a single input layer and dual convolution layers with three dense linear layers. To address overfitting, single max pooling is used to extract intensive features, single global average pooling, and a dropout layer. On the other hand, 2D CNN works with a single input layer followed by four convolution layers, dual max pooling to extract intensity features, three dropouts, and a flattened vector with dual dense classification layers. Bidirectional Long Short Term Memory is a specific Recurrent Neural Network type that works with the sequential data structure. The paper used a single input layer followed by 200 unit hidden and fully connected layer with a softmax activation function. For training purposes, feature surface variation and planarity decide the structure points to the confusion points to distinguish between the actual and predicted class label.

C. CLASSIFICATION OF DISASTER AFFECTED CULTURAL HERITAGE SITES

The long interval of the cultural heritage buildings are always history and thus suffer from various types of damages during the wage of the time. Nowadays, these methods are manual, and the operations are pretty time-consuming. Therefore, automation should mention informative changes in these fields by improving the classification performance of heritage versus non-heritage data with defective and non-defective classification. A multi-stage classification can normalize the field of study along with the collection of adequate data-driven algorithms.

[62] proposes a similar approach. Firstly, the data collection stage is taken place using social media data. There are two types of data that are being collected. Heritage, non-heritage building data, and defective, non-defective pallets are present in the heritage data. Secondly, social media data are primarily available with many outliers (here, wrong

search results). Manual effort can have a unique target by removing outlier data from the actual data. Therefore, data preprocessing and cleaning are essential steps to create the dataset needed for the classification task. Lastly, a Convolution Neural Network for the classification task is a good choice for the selected dataset and supervised class labels. Early detection of damage to cultural heritage because of disaster is a critical assessment. Kumar et al. classified images collected from social media to categorize whether an image has its main subject as cultural heritage and to categorize whether any damage is represented. Thus, automatic classification, though not perfect, helps detect images of damaged heritages to a great extent.

The paper also focuses on their approach to an extensive dataset collection to mature the crisis informatics domain by relying on time-dependent image structures. As a part of the noise cancellation process, they claim to discard 2,974 samples. They collected 6,612 heritage and 2,266 non-heritage samples and the damaged category of 836 heritage and 567 non-heritage samples from Twitter, google search, and video frames.

DL is an essential and effective parameter of the classification task. The classification model consists of two parts. Feature Extractor enables the power of extracting the good features from the image set according to the class labels using backpropagation. If the classification approaches deal mainly with segmented parts by classifying those into the proper categories, some well-known pretrained CNN feature extractors are used here. VGG16 [252], ResNet50 [59], DenseNet121 [26], InceptionResNetV2 [253], Xception [254], NASNet-Large [255] are used as a feature extractor, whereas, some of the ML approaches such as Logistic Regression [256], Support Vector Machine [257], Random Forest [247], AdaBoost [258] as an effective classifier.

The paper aims to work with two models named Heritage Model-1, representing heritage and non-heritage classification, followed by Heritage Model-2, differentiating damaged structures from normal ones. However, all the methods have been tested with different ML models on DL-based feature extractors on a case study on the Nepal Earthquake SMERP workshop dataset [259]. The paper presents a considerably good result for early defection classification as 90%.

D. NOVEL DATASET EXTRACTION AND ICONOGRAPHY CLASSIFICATION

The study of Iconography engages mental and logical experimentation based on the visual contents of a particular structure about its theme and motifs for easy characterization. Iconography is the discipline that studies and interprets graphic symbols and motifs present in an artwork. The importance of Iconography in Indian content represents the study behind heritage preservation based on shape, texture classification, and 3D reconstruction on a surface area of hyperplane based on the motifs. However, this approach is highly manual and thus needs ML algorithms' power to boost the mecha-

nism in a larger context. Therefore, classifying iconographic content plays a vital role in cultural heritage preservation as we need to identify the content similarity. This field is much less investigated and presents trifling challenges on structurally complex datasets of high-quality images. On the other hand, computer vision is a growing field of interest, engaging information retrieval, morphological analysis, clustering, and many more.

[63] presents a novel dataset of Iconography ArtDL followed a DL yet effective method towards the classification of to mark it as a content-based problem on art images. The dataset consists of 42,000 annotated images with 19 class labels of specific characters, e.g., Saint Dominic, Saint Jerome, and Saint John the Baptist. This dataset consists of 47,551 samples before filtering with a ratio of RGB (60%) and black-white (40%). Multiple duplicate images have been removed by calculating md5 has similarity and maintaining a certain threshold. In addition, imperfect brightness ratio and damaged, empty, fragmented images are removed using manual intervention in a time-consuming manner. Multiple augmentation technique is applied to normalize the dataset. Such augmentations are horizontal flip by 50%, normalization by the mean and standard deviation of the dataset, and random cropping. The dataset splitting contains 80% as the training set and 10% as the validation and testing set.

The classification task engages a DL algorithm as a backbone of the network. Here, ResNet50 [59] is used in a transfer learning approach to keep the knowledge of feature extraction trained on ImageNet [260]. To accomplish the classification chore on particular class labels in the dataset, the process removed the network's last layer, followed by a Global average pooling and neurons similar to the number of class labels. The first convolution layer, max-pooling, and first two residual blocks are being updated as it is needed to extract meaningful features from the new dataset. The architecture reflects an effective result with 72.73% Mean Average Precision followed by an Explainable AI technique to show specific gradients, which are used to indicate particular decisions made by the trained model. Class Activation Map [261] is one of the first XAI approaches mentioning the positive gradients to flow backward like a decision tree to activate maps responsible for making particular optimistic predictions to the images.

E. CLASSIFICATION OF ANCIENT MAYA GLYPH

The ancient heritage of Maya glyphs is primarily healthy and preserved and a severe field of interest due to atomic predictions world. Such glyph resides within a massive area of historical interest and thus needs to be kept based on understandability and readability perspectives. Apart from other scripts, the Maya glyph is visually attractive, maintaining the difference between each character similar to the modern languages. These also include some ancient animals as glyph characters identical to the hieroglyph. The first step of preservation is to classify different shapes and structures present in

the glyph so that sectioning and information retrieval can be done in the next step.

DL and ML are powerful tools to automate this effort rather than human intervention. [66] proposed a method to crowdsource the data in the first place to collect the dataset. Firstly, they take this dataset into a step of preprocessing for the oversampling task. They introduce several augmentations based on geometrical center transformation, e.g., rotation (within $-15, 15$ degrees), vertical and horizontal translation, random crop, and zoom ratio. 12 by keeping the data alteration limited. The paper started with ten glyphs for every ten types and propagated to 150 classes hereafter.

Secondly, this dataset can be used as the training set for different DL algorithms proposed in the paper. However, the paper focuses on experimenting with convolution neural network algorithms with three different setups.

- A standard pretrained feature extractor of VGG16 [252], ResNet18, ResNet50 [59] trained on ImageNet [260], Sketch-a-Net [262] pretrained on 250 class binary sketches followed by a shallow change in the last layer.
- The last convolution block is removed, followed by a dense layer ending with the number of classes present in the crowdsourcing dataset.
- Rather than taking pretrained features, train the full network from scratch.

Thirdly, the explainable AI technique GradCAM [263] is used to support classifier decisions. The classifier tends to be a black box that does not interpret the probability values given as a score to the classification task. GradCAM throws a heatmap to the image, propagating back through the whole map of the classification layer to the picture. It shows a significant amount of red area on which the model focuses to give the classification decision.

The Maya civilization used a writing system called *Maya glyphs*. Training any network for hieroglyph classification is challenging since segmented Maya glyphs are not available in plenty. Can et al. [66] make use of three different CNN architectures to train them for classifying hieroglyphs? The first approach uses a pre-trained CNN as a feature extractor. The second approach fine-tunes a pre-trained network. The third one considers a CNN from scratch. The paper concluded that a sketch-specific sequential network outperforms a generic sequential model (LeNet) and Residual Networks for the classification task.

VII. BUILDING INFORMATION MODELING

The pioneering effect of the “Building Model” was first introduced in the mid-1980s by Ruffle [68]. The paper proposed the first digital concept of intelligent building modeling, followed by various automated approaches. However, the term “Building Information Modeling” comes after ten years with some informative data extraction using computerized approaches into action [264], [265], [266]. Such methods are still helpful in collecting data from cultural heritage buildings and informative modeling of monumental structures. These

approaches help to localize the lack of importance in digital information, followed by rephotography, photogrammetry, and manual hovering-based approaches using computer vision. The cloud storage of the data is assembled with the massive availability of the internet. Creating and managing information regarding a built asset is known as Building Information Modeling (BIM) [70]. Scan to BIM methodology is used for 3D reconstruction from point clouds obtained by LiDAR and PH. We can use this process to create models where there is a lack of information in input data [71].

A. SEMIAUTOMATIC BIM ON SEMANTIC POINT CLOUD EXTRACTION

A method on point cloud extraction followed by building information modeling in [72] presents informative criteria for reconstructing rooftop-based architectures. The work proposes a semi-automatic approach in the context of heritage 3D reconstruction rather than 2D as it explores more viable options with a better visualization. The key features of this method are as follows:

- The first step introduces photogrammetry and laser scanning for the fast collection of heritage point cloud data. However, the data collection is mostly unstructured, thus requiring human intervention for data preprocessing (e.g., registration of point cloud, filtering, noise reduction). This data is documented by uploading it to various cloud platforms.
- Secondly, the paper uses ML techniques to identify primitive geometrical features from the data clouds using neighborhood selection methods (e.g., KNN, clustering techniques). To create the training data, such features are extracted in convolution feature vectors followed by a manual annotation with supervised labels. The method then forwards this label to the Random Forest [247] classifier in MATLAB, which learns to annotate a test building.
- The annotated ML model conveys a multi-label classification with different color shades. A particular label-based extraction can motivate parametric joint correction by geometrical correction algorithms to reconstruct visually absent parametric components of the 3D model point clouds. This label correction is then added using coordinate fitting to the actual picture to reconstruct the absent area on the picture.

B. BUILDING DIGITAL TWIN FROM BUILDING INFORMATION MODELING

Cultural Heritage buildings are mostly old and historical and thus need up-to-date checking based on their quality assessment and endangerment criteria. Building Digital Twin (BDT) is motivated by advances of BIM that enforce an updated overview of methodologies using computer vision. This approach mentions time-based framing using various object detection and localized approaches on 2D to 3D transformation rather than employing point clouds directly. Therefore, keeping aside the point cloud extraction, this process is competitively faster. The recent advancement of BDT comes with IoT techniques depending on several conditions. The synchronization of the outside world from CCTV based

network is quite challenging and time-consuming due to various reasons. One of the first of these is Dimension Inconsistency by capturing 2D geometry from the image, keeping the depth apart. Dimension Inconsistency results in Coordinate Inconsistency that plots physical structure differently from CCTV images. The very last step of it is due to the previous two inconsistency that results in Object inconsistency, making wrong geometrical predictions of object structures and depth maps.

Therefore, the recent advancement of Computer Vision is a viable solution that deals with a higher understanding of machines from Image and Video frames. This approach can be made in three steps. Firstly, a camera can capture the 2D geometry of an object or video feed of cultural heritage data. This image is then passed to object detection in two following steps. 1) 2D object detection using YOLOV5 [73], 2) 3D object estimation motivated from Pose Detection algorithm [267] using Feature vector of Multi-Layer Perceptron [268]. They further discuss Camera to BIM location transformation using the feature vector location to get a comprehensive idea about the image features to plot it into 3D coordinates.

$$\arg \max_{\mathcal{M}} \sum_{i \in I} \text{dist}(i, \mathcal{M}(i)), \quad \text{s.t., } \forall i, j \in I, \mathcal{M}(i) \neq \mathcal{M}(j) \quad (7)$$

The loss predicted as the maximum distance *dist* between the index value *i* and coordinate location extracted from the BIM $\mathcal{M}(i)$, where the two locations are induced from two different points and thus not similar $\mathcal{M}(i) \neq \mathcal{M}(j)$.

Secondly, an object matching mechanism is introduced using Cold Start object matching and Run Time object matching that calculates the trajectory of the object based on the previous ones, making it a consecutively linear space to the next steps to take a better decision. The cold start matching between the object present in the heritage site and the BIM model in case of camera calibration can be defined as equation 7.

C. A COMPARATIVE STUDY OF 3D IDEAL MODEL AND SCAN-TO-BIM

A comparative study between two well-known methodologies is essential to discuss the shortcomings and generate mixed literature for future endeavors. The patriarchy of 3D modeling directly connects 3D methods and cameras using photogrammetric approaches. On the other hand, SCAN-TO-BIM [69] utilizes the effect of feature extraction using point cloud data followed by ML approaches. Both methods are well documented with advanced automated models, thus effectively qualifying the user's interest. Capone and Lanzara have discussed a similar generation of the parametric 3D model from the point cloud in [269]. The dome of *S. Maria di Constantinopoli Church* has been generated using geometric rules. The objective is to analyze parametric modeling from point clouds against traditional 3D modeling for cultural

heritage studies. Though the approaches are reliable, these are time-consuming and waste big data.

In 3D modeling, four techniques are most important for point cloud analysis. The methods and their contributions are mentioned below:

- Reverse Modeling uses geometrical transformations to convert the point cloud into a polygon mesh surface structure.
- Direct Modeling transforms the tiny absorption on the surface area to reconstruct complex shapes.
- Generative Modeling captures the features from a large number of samples to generate a new figure belonging to the sample distribution.
- Parametric Modeling generally works with pre-modeled statistical features for BIM and HBIM analysis models.

For extensive data interpretation, there can be multiple steps that may be useful. The data acquisition stage is mainly part of the 3D scanning point cloud using laser scanning or photogrammetry. Some well-known parametric models with the idea of reality-based models are used to comprehend critical featured data and generate complex parametric models that enforce 3D communication approaches within HBIM and 3D modeling.

VIII. VIRTUAL REALITY SYSTEMS

This category includes virtual, augmented and mixed reality experience of architectural structures. It also explores the use of haptic feedback mechanism which is more cost effective compared to the 3D printed replicas of artifacts.

Virtual reality (VR) or immersive reality is a computer-based simulation of the real world. The users may interact with the created environment and feel completely immersed within it. A virtual walk through around any museum serves as an application of VR. Whereas, augmented reality (AR) is a technology for superimposing additional digital content on real world environment for enhancing the experience of reality. And, mixed-reality is a combination of both VR and AR. Immersiveness is represented by a continuous scale known as reality-virtuality continuum which ranges from real to completely virtual environments.

Different concepts and methodologies in the domain of virtual restoration and virtual reconstruction has been discussed by Pietroni and Ferdani in [74]. It is opined that digital replication of original cultural heritage helps in understanding past culture better. *Teramo Virtual City*, *The Regolini Galassi Tomb*, *The Roman Villa of Aiano* etc. are some of the 3D reconstructions studied here.

3D cultural heritage artifacts can be well inspected by interactive virtual reality (VR) methods [75]. Also we see here that haptic feedback enhances virtual experiencing of 3D objects for assessment of object shape.

An evaluation on the ability of a haptic interface enables to understand and improve the experience of users with virtual museum systems. A haptic application of the *Museo Archeologico Nazionale delle Marche*, based on Omega.6, a force-feedback device has been focussed in [270] by Ceccacci et al. Comparison between haptic and mouse-based interfaces are

highlighted. Also, gamification which is another strategy to convey information about the museum to visitors in an interactive manner has been discussed.

Rodriguez et al. [76] have explored 3D representation of museum artifacts for enhancing the virtual experience of visitors with the help of interactive systems. The *Mont'e Prama Complex*, which is an extraordinary protostoric Mediterranean sculpture collection is depicted digitally using an interactive system. An analysis has been carried out based on the feedback of thousands of visitors on their preferences and behaviours during museum exploration.

An approach for perceiving the digital collection of sculptures of the museum through an indirect touch-based interactive control of a large display surface has been presented in [271]. Also the behaviours of visitors during interactive explorations have been computed from histogram analysis of view parameters and cluster analysis methods.

Cisternino et al. in [272] presented how augmented and spatial augmented reality technologies can be used for exploring historical monuments such as the Basilica of Saint Catherina, Alexandria, located at Galatina. These approaches enable visitors to know about the infinite facets of a monument easily. A questionnaire on the experience of museum visit is also presented.

Augmented reality and gigapixel image technology has been used in [273] by Gonzalez-Lopez et al. to achieve conservation and restoration of an Andalusian Baroque altarpiece.

Yi and Sun characterized museum experience step wise regarding mixed reality(MR) technology using Hololens in [274]. Four types of museum experiences namely objective, cognitive, introspective and social experiences are defined. Museum viewing experience of visitors are evaluated on the responses of user surveys. Thus essential factors of MR experiences and six types of user experiences viz. knowing, restoring, exploring, expanded scale, encountering and sharing have been categorized.

Traditional art venues and heritage architectures have been made accessible to visually impaired people with the help of a ring known as *tooteko*. The *tooteko* when applied over any tactile surface tagged with NFC sensors, returns an audio track corresponding to portion of the surface touched [275].

A survey on the exploration, reconstruction and exhibition of cultural heritage using virtual, augmented and mixed realities has been presented by Bekele et al. in [276]. A list of applications of VR technology has also been tabulated here.

Digital mobile guides of museums is a developing practice of today. However, in most cases, digital guides fail to provide personalized approaches that can be provided by human guides. To address this, Antoniou et al. [277] explored how digital tools can be used by human guides to assist them in guiding visitors. A human-led hybrid digital approach for experiencing museums has been proposed here which links guiding experience and rich cultural databases supported by their related narratives.

IX. OTHER DIGITAL RESTORATION TECHNIQUES

A. GENETIC ALGORITHM

The evergreen Darwinian theory of Natural Evolution introduces the notion of selection of the fittest from a finite population. Natural selection supports selection, mentioning various approaches of unsupervised heuristic search algorithms. Such approaches term the mixing as the crossover (recombination) of features (biological chromosomes) to create a better function based on a fitness function (selection criteria). Therefore, genetic approaches to search and optimize the problem of object and shape reconstruction from fragments followed by natural selection can be a specific application in real-world scenarios. However, various 3D modeling and supervised DL reconstructions have already gone through a set of improvements. However, they need help with more extensive reconstruction of real-world wall paintings in time dependency and considerable results. Local assembly of object fragments to reconstruct the complete geometry of the architecture is already addressed in various articles [278], [279] by finding the adjacency. On the other hand, Global Assembly focuses on searching the fragment space to find particular pairwise segments by rotation and positional orientation to solve the local assembly problem in a larger space.

Wall Painting Reconstruction has almost become a classical problem in heritage restoration. Such artworks are essential for future generations and thus need to be preserved. However, many artworks are already on the verge of collapse and thus need scientific exploration to reconstruct their history. Though several DL approaches are useful [77] for small test cases, significant problems remain in the darker region, which needs to be explored soon.

The genetic algorithm can solve the larger context for this task. [78] presents a novel unsupervised genetic approach for wall painting reconstruction by updating the point of partial reconstruction for the recombination based on fitness function solved by evolutionary computation. The approach uses an iterative eigenvector technique that is resilient to detecting outliers near the partial reconstruction point. Such corrections are helpful for incorrect shape matching and position encoding format. Stating individual fragments as a cluster, the probabilistic procedure is held accountable for making new best samples in the presence of parent clusters. This method enforces recombination by matching fragment clusters followed by a merging. After creating the best samples, the new clusters go through a feasibility check. Let $MaxST(C_i)$ be the sum of the match scores of the maximal spanning tree of cluster C_i . Then the fitness function of the cluster C_i is

$$f(c_i) = MaxST(C_i) - W(span_{f_i} + span_{m_i}) \quad (8)$$

A convex optimization method is used as a fitness function equation 8, then ranks the clusters and removes the outliers based on selection criteria. Top-rank clusters are selected for the subsequent mixing that can give a better visualization of wall paintings.

Another approach [280] deploys a genetic algorithm along with hill climbing followed by image features from the

dataset. This article aims to reconstruct artifacts by combining the broken pieces and maintaining the correct positions. In this paper, the author uses a hill-climbing algorithm to correct the random selection of samples from the population based on a fitness function. The image designs the fitness function features using Oriented Fast and Rotated Brief(ORB), which uses a positional encoding of objects to correct the position of two fragments using color-wise matching. The slope of two similar points remains zero keeping the different ones to optimize using a hill-climbing algorithm. After one step, the genetic algorithm selects the best samples from the population, using the optimization from the fitness function, which uses ORB image features.

B. PROBABILISTIC METHODS

We obtain cultural relics mostly in fragments, which need reassembly to make them worthwhile for cultural heritage studies. However, automatic reassembling fragments is not very easy as the geometric shapes of the fragments may not be known. A novel probabilistic method for reassembling fragments has been proposed by Wang et al. in [281]. In this method, 2D link-chain and 3D spatial-description descriptors are used to detect matching units of the fragments, and then the reassembly probability pairwise is calculated. We demonstrate the efficiency of the method using eight challenging datasets.

Data from point clouds of historical places are extracted and analysed using a method called probabilistic modeling-based point cloud extraction. It entails precisely identifying and categorising various point cloud data items, such as structures, trees, and terrain, using statistical techniques and machine learning algorithms. This method makes it possible to analyse historical sites more effectively and accurately, which can aid in preservation efforts, archaeological study, and other related sectors. Researchers can better comprehend the location and its history by evaluating the structure and makeup of the point cloud data, which will help them make better judgements about how to conserve and analyse it.

1) AUTOMATIC RECONSTRUCTION OF HERITAGE SITE FROM LiDAR DATA

A new probabilistic approach for reconstructing building roofs from airborne LiDAR data is demonstrated in [282]. By using elevation and point density to identify probable roof points, the proposed technique then generates a three-dimensional roof model by fitting a probabilistic plane model to the determined roof points. The method was evaluated on three various datasets and yielded promising results when compared to other cutting-edge techniques.

2) POSE ESTIMATION AND PROBABILISTIC 3D OBJECT RECOGNITION

In this study [283], point cloud data is used to suggest a probabilistic method for 3D object detection and posture estimation in cluttered situations. To estimate the object's

posture, the method combines a probabilistic model of the scene and a probabilistic model of the object's shape and appearance. The method was tested using real-world datasets, and even in congested settings with occlusions and partial views, it produced good results.

C. INTERNET OF THINGS (IoT) AND ARTIFICIAL INTELLIGENCE (AI)

Restoration of cultural heritage is a critical aspect of preserving cultural assets. One important aspect for preserving the heritage data is to use cloud computing based platforms [284]. An intelligent sensor [285], [286], [287] is also an efficient tool for collecting and monitoring data from objects or sites related to cultural heritage. This data can be further analyzed with the help of AI. Talamo et al. described smart portable sensors for detecting anomalies and risk management in [288]. Based on the data received from low-cost wireless sensors allocated at target areas, a risk prediction system has been designed to predict the long-term effect of the surrounding environment on the target cultural asset. Experiments have been carried out through sensors located in three rooms of *Uffizi Gallery* in Florence to predict the effect of air pollution on the gallery rooms.

A recurrent neural network empirically evaluated with recurrent units, which includes a long short-term memory(LSTM), and Woosik and Dong-hoon in [289] proposed a gated recurrent unit(GRU). A CNN-LSTM and CNN-GRU type hybrid model was developed to analyze the time series sensor data obtained from sensors installed on the Woljeong Bridge, South Korea, to predict any damage or weathering of the bridge. The proposed method uses IoT to collect data from the bridge continuously.

In the construction sector, the adoption of AI has several benefits as well as some disadvantages. Goussous et al. detailed the possibilities and impacts of using AI in the construction sector in [290]. The use of AI for the restoration of *Petra*, a world-famous heritage site in Jordan, is studied. It could be concluded that using AI enhances the pace of restoration and is also more cost-effective.

X. POTENTIAL FUTURE RESEARCH DIRECTIONS

The future compass of this paper depends upon the pertinence of various procedures on cultural heritage data. We accentuate the commission of mechanical learning-based restoration procedures instead of manual intervention. Such methods preserve time by recreating a mathematical function to accomplish identical tasks. Regardless, most learning-based approaches require large-scale data to train the model, maintaining intuitively good results for small-scale test cases. Therefore, numerous automated procedures have been presented on diverse topics [66], [269] to sustain and elevate the restoration of heritage images. A plethora of datasets [21], [22], [23], [58] and the Ducal Palace dataset in Urbino, Palazzo Ferretti in Ancona, Italy conveys the training of more additional learning-based algorithms for different tasks. This paper discusses dataset acquisition, followed by the 3D

reconstruction strategy. 3D reconstruction mainly focuses on reconstructing the whole structure or identifying the area of damage within the architecture. Digital inpainting methods focus on painting the area of the damage in the architecture followed by classification approaches [47], [58]. Numerous techniques with manual statistical fitness functions are set up to evaluate this task's genetic algorithms.

Although the last five years of research on "Digital Restoration" has outstretched in an impressive state using Computerized tools and methods, but still yields ethnicity because of potentiality. Below we unfold some potential future research directions that can be valuable.

- 1) The initial stage of heritage restoration mainly concentrates on the 3D reconstruction of heritage constructions. Point Cloud extraction can be an important facet of computerizing the manual measure. After the automated extraction, [47] proposed a texture-based segmentation on point cloud data. Rather than using a manual statistical approach, we allude to the conjecture from [29] that proposes Graph Convolution operation keeping it computationally costly and time-consuming to drill the model. The point cloud extraction is optimized and can be applied on a large scale using [50]. The right extraction from the image supports the locality of reference of the image. The 3D locality can be stalked by a 2D object detection algorithm like YOLO [73] and mapped to the segmentation distinction between point clouds having different localities. This methodology rescues much computation in segmenting point clouds, where we take help from 2D objects to rank them in a 3D point cloud. This point cloud can then reconstruct the object utilizing software-based tools.
- 2) The improvement in the second stage aims to replace the software-based reconstruction strategies using IOT detectors. The Internet Of Things fuses advanced sensors into one network, considering high-speed internet connectivity inclined. Such sensors can grab the geometry of the structure nearby. After this induction and operation, we can use the components of the architecture to reassemble it into a fully structured 3D image using DL approaches [291], [292], [293]. Employing this work in the digital heritage reconstruction could be a practical, time-saving approach instead of applying software. After 3D reconstruction, Identity-based mapping can be enforced using a classification and point prognosis approach from the position of 3D sensors.
- 3) One of the vital approaches using DL can be the GAN-based digital inpainting process to seal the internal voids based on the story mentioned in the nearby area. Story Based GAN [294] seizes the time sequence story mentioned in the previous segments to forge a new sample based on stories. We set the training dataset in a series of tiles that mentions a story. Thus, delivering the succession of tiles in the trained model can render

a new tile that can be used to fill the area of damage associated with digital inpainting. To revamp a 3D model of the 2D projection of a new tile, we employ DL scenarios [25].

- 4) The empowerment of computer vision generates noteworthy data based on the training set. In the prior direction, we utilize story-based GAN architecture to fine-tune the interpretation of a local variance shift using time series prediction. On the other hand, diffusion models are creating a new age resulting in real-life HD pictures [295] and 3D diffusion [296]. Examining the StoryGAN in diffusion-based DreamFusion architecture will change certitude by laborsaving inpainting from stories mentioned in the tiles, followed by high-quality 3D reconstruction of the heritage buildings. Such methods could also reproduce the viewpoint of real-time inpainting using video inpainting [297] that restores the broken tiles of a building.

XI. CONCLUSION

The restoration of digital heritage is mesmerizing, which guides to recreating history using futuristic plans. The article strives to explain and discuss some well-known works that can be applied to heritage restoration. We infer 3D reconstruction as an essential task pursued by various classes based on complexity for restoring broken furs of inheritance. Nonetheless, mechanization in the field still holds numerous unanswered questions. Hence, first, we chose to commence with some of the well-known resolvable scenarios using some of the popular dataset collection methods [6], [69]. Such methods result in unique datasets, e.g., [21], [22], [23], [58], which can be used for self-regulating training approaches even further. We use DL and image processing approaches to mark their capability on artistic legacy, followed by GAN, BIM, and discriminating case studies of some popular architectures worldwide. This paper presents several holistic approaches to data collection, computer vision, point cloud, and genetic algorithm-based segmented analysis of heritage image data. We then proceed with an exhaustive analysis of the papers in this survey and conclude with potential research direction on restoring cultural heritages.

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